

APPLIED ECONOMICS

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Chapter 10: Linear Programming**1. Introduction**

The theory of linear programming provides a good introduction to the study of constrained maximization (and minimization) problems where some or all of the constraints are in the form of inequalities rather than equalities. Many models in economics can be expressed as inequality constrained optimization problems. A linear program is a special case of this general class of problems where both the objective function and the constraint functions are linear in the decision variables.

Linear programming problems are important for a number of reasons:

- Many general constrained optimization problems can be approximated by a linear program.
- The mathematical prerequisites for studying linear programming are minimal; only a knowledge of matrix algebra is required.
- Linear programming theory provides a good introduction to the theory of duality in nonlinear programming.

Linear programs appear in many economic contexts but the exact form of the problems varies across applications. We shall present several equivalent formulations of the basic linear programming problem in this introductory section. In the following section, we provide a geometric interpretation of a linear program (LP) in activities space. In subsequent sections, we will present George Dantzig's (1963) *simplex algorithm* for solving an LP.¹

Our *first formulation* of the basic linear programming problem is:

$$(1) \min_x \{c^T x : Ax = b ; x \geq 0_N\}$$

where $c^T = [c_1, c_2, \dots, c_N]$ and $b^T = [b_1, b_2, \dots, b_M]$ are N and M dimensional vectors of constants, $A = [a_{mn}]$ is an M by N matrix of constants and $x^T = [x_1, x_2, \dots, x_N]$ is a *nonnegative* N dimensional vector of decision or choice variables. Thus this first form for a linear programming problem is the problem of minimizing a linear function $c^T x$ in the vector of nonnegative variables $x \geq 0_N$ subject to M linear equality constraints, which are written in the form $Ax = b$.

¹ "Linear programming was developed by George B. Dantzig in 1947 as a technique for planning the diversified activities of the U.S. Air Force." Robert Dorfman, Paul A. Samuelson and Robert M. Solow (1958; 3). Dorfman, Samuelson and Solow go on to note that Dantzig's fundamental paper was circulated privately for several years and finally published as Dantzig (1951). A complete listing of Dantzig's early contributions to developing the theory of linear programming can be found in Dantzig (1963; 597-597).

Our *second formulation* of an LP is:

$$(2) \max_{x_0, x} \{x_0 : x_0 + c^T x = 0 ; Ax = b ; x \geq 0_N\}$$

where x_0 is a new scalar variable, which is defined by the equality constraint in (2); i.e., $x_0 \equiv -c^T x$ and so minimizing $c^T x$ is equivalent to maximizing x_0 . It can be seen that the first and second formulations of an LP are completely equivalent.

Our *third formulation* of an LP is the following problem:

$$(3) \max_x \{c^T x : Ax \leq b ; x \geq 0_N\}.$$

The above problem can be transformed into a problem of the type defined by (2) as follows:

$$(4) \max_{x_0, x, s} \{x_0 : x_0 - c^T x = 0 ; Ax + I_M s = b ; x \geq 0_N ; s \geq 0_M\}$$

where I_M is the M by M identity matrix. Note that we have converted the inequality constraints $Ax \leq b$ into equality constraints by adding an M dimensional vector of nonnegative *slack variables* s to Ax .

Note that equality constraints such as $Ax = b$ can be converted into inequality constraints by writing $Ax = b$ as two sets of inequality constraints: $Ax \leq b$ and $-Ax \leq -b$. Note also that it does not matter whether we are maximizing the objective function $c^T x$ or minimizing $c^T x$ since a problem involving the maximization of $c^T x$ is equivalent to a problem involving the minimization of $-c^T x$.

In the above problems, the vectors of variables x and s were restricted to be nonnegative. This is not an essential restriction, since if a variable, x_1 say, is allowed to be unrestricted, it may be written in terms of two nonnegative variables, say s_1 and s_2 , as $x_1 = s_1 - s_2$. However, in most economic problems, restricting the decision variables x to be nonnegative will be a reasonable assumption and it will not be necessary to resort to the $x_1 = s_1 - s_2$ construction.

Thus the essence of a *linear programming problem* is that the objective function (the function that we are maximizing or minimizing) is *linear* and the constraint functions are *linear equalities or inequalities*.

It turns out that our second formulation of an LP is the most convenient one for actually solving an LP (the simplex algorithm due to Dantzig) but the third formulation is the most convenient one for proving the duality theorem for linear programs.

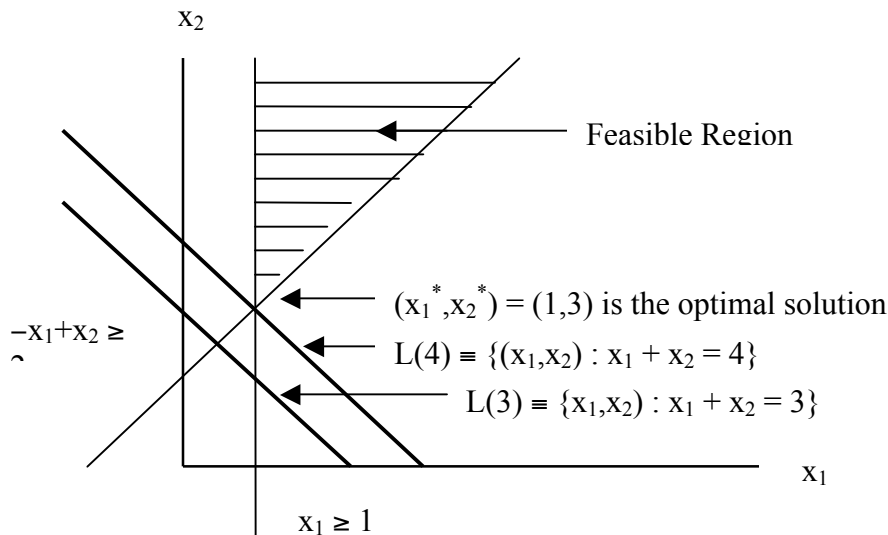
2. The Geometric Interpretation of a Linear Program in Activities Space

Consider the following LP:

$$(5) \min_x \{c^T x : Ax \geq b; x \geq 0\}$$

where $c^T \equiv [1,1]$, $b^T \equiv [1,2]$ and the two rows of A are $A_1 \equiv [1,0]$ and $A_2 \equiv [-1,1]$. Thus we wish to minimize $x_1 + x_2$ subject to the four inequality constraints: $x_1 \geq 1$; $-x_1 + x_2 \geq 2$; $x_1 \geq 0$; and $x_2 \geq 0$. We define the constraint set or *feasible region* in x space or *activities space* to be the set of x 's which satisfy the constraints in (5). It is the shaded set in Figure 1 below. We also graph the level sets of the objective function $x_1 + x_2$; i.e., these are the family of straight lines indexed by k , $L(k) \equiv \{x : c^T x = k\}$. These level sets form a system of parallel straight lines. Obviously, the optimal x solution to the LP problem will be that x which belongs to the feasible region and which also belongs to the lowest level set of the objective function.

Figure 1: A Linear Program in Activities Space



Note that the optimal solution to the LP lies on the boundary of the feasible region and that the optimal level set of the objective function is tangent to the feasible region. Finally, note that the optimal solution to the LP is at a *vertex* of the feasible region. This is typically what happens.

3. The Simplex Algorithm for Solving Linear Programs

In this section, we outline Dantzig's (1963; chapters 5-7) simplex algorithm for solving linear programming problems.² Dantzig's method is not only of interest from a computational point of view, but also from a theoretical point of view, since it enables us

² Actually, we present a version of Dantzig's (1963; chapter 9) revised simplex algorithm.

to present an entirely algebraic proof of the duality theorem for linear programming problems as we shall see later. However, before the method can be explained, we need some additional definitions and results.

Definition 1: Any solution x^0 to the system of linear equations $Ax = b$ and inequalities $x \geq 0_N$ is called a *feasible solution* to the LP defined by (2).

We are considering the system of linear equations, $Ax = b$, which occurs in (2) above where A is an M by N matrix. In what follows, we assume that the number of equations M is less than the number of x_n variables, which is N . Hence A is a singular matrix.

Definition 2: Any nonnegative solution $x^0 \geq 0_N$ to the system of equations $Ax = b$ with at least $N-M$ components equal to zero is called a *basic feasible solution* to the LP defined by (2).

Theorem 1; Carathéodory (1911; 200), Fenchel (1953; 37), Dantzig (1963; 113-114): If $Ax = b$, $x \geq 0_N$ has a feasible solution, then it has a basic feasible solution.

Proof: Assume that $M < N$ and that there exists an $x^* = [x_1^*, \dots, x_N^*]^T \geq 0_N$ such that $Ax^* = b$. If x^* has M or less nonzero components, then x^* is a basic feasible solution and we are done. Thus we assume that x^* has $K > M$ nonzero components. By reordering the variables if necessary, we assume that the first K components of x^* are the nonzero components. Now look at the following system of equations:

$$(6) \sum_{k=1}^K A_{\bullet k} \lambda_k = 0_M$$

where $A_{\bullet k}$ is the k th column of the A matrix. The matrix $[A_{\bullet 1}, A_{\bullet 2}, \dots, A_{\bullet K}]$ is M by K where $K > M$ and hence this matrix is singular or alternatively, the K columns in this matrix are linearly dependent. Hence, there is a nonzero $\lambda^* = [\lambda_1^*, \dots, \lambda_K^*]^T$ solution to (6). Without loss of generality, we can assume that at least one component of λ^* is positive.³ We also have $x_k^* > 0$ for $k = 1, 2, \dots, K$. Hence the number α defined by (7) below must be positive:

$$(7) \alpha \equiv \max_k \{ \lambda_k^* / x_k^* : k = 1, 2, \dots, K \} > 0.$$

Note that:

$$(8) \alpha \geq \lambda_k^* / x_k^* \text{ or } \alpha x_k^* \geq \lambda_k^* \text{ for } k = 1, 2, \dots, K.$$

Since $x^* \geq 0_N$ is a feasible solution for $Ax = b$, we have:

$$\begin{aligned} (9) \quad b &= Ax^* \\ &= \sum_{k=1}^K A_{\bullet k} x_k^* \\ &= \sum_{k=1}^K A_{\bullet k} x_k^* - \alpha \sum_{k=1}^K A_{\bullet k} \lambda_k^* \end{aligned} \quad \text{using (6)}$$

³ If all components of λ^* were 0 or negative, then $-\lambda^*$ would satisfy (6), with all components 0 or positive.

$$= \alpha^{-1} \sum_{k=1}^K A_{\bullet k} [\alpha x_k^* - \lambda_k^*].$$

But for all k which attain the max in (7), we will have $\alpha x_k^* - \lambda_k^*$ equal to 0. Thus $y_k^* \equiv \alpha^{-1}[\alpha x_k^* - \lambda_k^*]$ for $k = 1, 2, \dots, K$ is a feasible solution for $Ax = b$ with at least one additional zero component compared to our initial feasible solution x^* .

We can continue this process of creating extra zero components as long as K , the number of nonzero variables, exceeds M , the number of equations. The procedure may stop when $K = M$, but at that stage, we have a basic feasible solution to $Ax = b$. Q.E.D.

In order to initiate the simplex algorithm, we need to start the algorithm with a basic feasible solution. But the above results say that given an arbitrary feasible solution, we can construct a basic feasible solution in a finite number of steps and thus initiate the simplex algorithm.

We can rewrite the basic linear program defined by (2) above as follows:

$$(10) \max_{x_0, x_1 \geq 0, x_2 \geq 0, \dots, x_N \geq 0} \{x_0 : e_0 x_0 + \sum_{n=1}^N A_{\bullet n}^* x_n = b^*\}$$

where $c^T \equiv [c_1, \dots, c_N]$, $A \equiv [A_{\bullet 1}, A_{\bullet 2}, \dots, A_{\bullet N}]$ is the original A matrix and

$$(11) A_{\bullet n}^* \equiv \begin{bmatrix} c_n \\ A_{\bullet n} \end{bmatrix} \text{ for } n = 1, \dots, N, b^* \equiv \begin{bmatrix} 0 \\ b \end{bmatrix} \text{ and } e_0 \equiv \begin{bmatrix} 1 \\ 0_M \end{bmatrix}.$$

In what follows, we also need to define the following M *unit vectors* of dimension $M+1$:

$$(12) e_1^T \equiv [0, 1, 0, \dots, 0] ; e_2^T \equiv [0, 0, 1, 0, \dots, 0] ; \dots ; e_M^T \equiv [0, 0, 0, \dots, 1] .$$

Thus e_m is an $M+1$ dimensional vector consisting of 0 elements except entry $m+1$ is 1.

Assume that a feasible solution to (10) exists. Then by Theorem 1, a basic feasible solution exists. By relabelling variables if necessary, we can assume that the first M columns of A correspond to the nonzero variables in this basic feasible solution. Thus we have $x_0^0, x_1^0 \geq 0, x_2^0 \geq 0, \dots, x_M^0 \geq 0$ such that:

$$(13) e_0 x_0^0 + \sum_{m=1}^M A_{\bullet m}^* x_m^0 = b^* .$$

At this point, we make two additional assumptions that will be relaxed at a later stage.

Nonsingular Basis Matrix Assumption: $[e_0, A_{\bullet 1}^*, A_{\bullet 2}^*, \dots, A_{\bullet M}^*]^{-1} \equiv B^{-1}$ exists.

Nondegenerate Basis Matrix Assumption: $x_1^0 > 0, x_2^0 > 0, \dots, x_M^0 > 0$; i.e., the x_m^0 which satisfy (13) for $m = 1, 2, \dots, M$ are all positive.

We define the initial $M+1$ by $M+1$ basis matrix B as $B = [e_0, A_{\cdot 1}^*, A_{\cdot 2}^*, \dots, A_{\cdot M}^*]$. By the nonsingular basis matrix assumption, we have the existence of B^{-1} and

$$(14) B^{-1}B = B^{-1}[e_0, A_{\cdot 1}^*, A_{\cdot 2}^*, \dots, A_{\cdot M}^*] = I_{M+1} = [e_0, e_1, \dots, e_M].$$

Now premultiply both sides of the constraint equations in (10) by B^{-1} . In view of (14), the resulting system of $M+1$ equations is the following one:

$$(15) e_0x_0 + e_1x_1 + e_2x_2 + \dots + e_Mx_M + B^{-1}A_{\cdot M+1}^*x_{M+1} + B^{-1}A_{\cdot M+2}^*x_{M+2} + \dots + B^{-1}A_{\cdot N}^*x_N = B^{-1}b^*.$$

If we set $x_{M+1} = 0, x_{M+2} = 0, \dots, x_N = 0$ in (15), then we obtain our initial basic feasible solution to the LP problem:

$$(16) x_m^0 = B_{m\cdot}^{-1}b^* > 0 \text{ for } m = 1, 2, \dots, M$$

where $B_{m\cdot}^{-1}$ is row $m+1$ of B^{-1} and the strict inequalities in (16) follow from the nondegenerate basis matrix assumption.

The question we now ask is: is our initial basic feasible solution to the LP (10) the optimal solution to the problem or not? To answer this question, look at equations (15) with $x_{M+1}, x_{M+2}, \dots, x_N$ all equal to zero. If we try to increase any of these last $N-M$ x_n variables from its initial 0 level, then obviously, we will increase x_0 only if $B_{0\cdot}^{-1}A_{\cdot n}^* < 0$ for some $n > M$. Thus we have the following:

(17) *Optimality Criterion:* If $B_{0\cdot}^{-1}A_{\cdot n}^* \geq 0$ for $n = M+1, M+2, \dots, N$, then the current basis matrix and the corresponding solution (16) solve the LP (10). If $B_{0\cdot}^{-1}A_{\cdot n}^* > 0$ for $n = M+1, M+2, \dots, N$, then the current basis matrix and the corresponding solution provide the unique solution to the LP (10).

Since $B_{0\cdot}^{-1}A_{\cdot n}^* = 0$ for $n = 1, 2, \dots, M$ using (14), the first part of the optimality criterion can be changed to:

$$(18) B_{0\cdot}^{-1}A_{\cdot n}^* \geq 0 \text{ for } n = 1, 2, \dots, N.$$

Suppose now that column s (not in the basis matrix) is such that

$$(19) B_{0\cdot}^{-1}A_{\cdot s}^* < 0.$$

In view of the nondegeneracy assumption, it can be seen that we can increase the value of our objective function x_0 and satisfy the inequality constraints in the LP (10) by increasing x_s from its initial 0 level. The question now is: which column should be dropped from the initial basis matrix B ? To answer this question, look at the following system of equations:

$$(20) \quad e_0 x_0 + e_1 x_1 + e_2 x_2 + \dots + e_M x_M + B^{-1} A_{\cdot s}^* x_s = B^{-1} b^*.$$

Suppose that $B_{m\cdot}^{-1} A_{\cdot s}^* \leq 0$ for $m = 1, 2, \dots, M$. Then as we increase x_s from its initial 0 level, the last M components of $B^{-1} A_{\cdot s}^*$ are either 0 or negative and these nonpositive numbers can be offset by increasing $x_1, x_2, \dots, x_M$ from their initial values $x_1^0, x_2^0, \dots, x_M^0$. Thus we have a feasible solution to (10) which will allow the objective function to attain any large positive value. In this case, we obtain the following:

(21) *Unbounded Solution Criterion*: If we have a column s such that $B_{0\cdot}^{-1} A_{\cdot s}^* < 0$ and $B_{m\cdot}^{-1} A_{\cdot s}^* \leq 0$ for $m = 1, 2, \dots, M$, then we may increase x_s to $+\infty$ and obtain an unbounded solution to the LP (10).

Suppose now that (19) holds but that $B_{m\cdot}^{-1} A_{\cdot s}^* > 0$ for at least one index $m \geq 1$. In this case, as we *increase* x_s from its initial 0 level, in order to satisfy equations (20), we must *decrease* x_m from its initial positive level $x_m^0 = B_{m\cdot}^{-1} b^* > 0$. Thus an upper bound to the amount that we can increase x_s before we violate the nonnegativity constraints $x \geq 0_N$ is $B_{m\cdot}^{-1} b^* / B_{m\cdot}^{-1} A_{\cdot s}^*$. Now take the minimum of all such upper bounds over all indexes m such that $B_{m\cdot}^{-1} A_{\cdot s}^* > 0$:

$$(22) \quad x_s^* \equiv \min_m \{B_{m\cdot}^{-1} b^* / B_{m\cdot}^{-1} A_{\cdot s}^* : m \geq 1 \text{ and } m \text{ is such that } B_{m\cdot}^{-1} A_{\cdot s}^* > 0\}.$$

The algebra in the above paragraph can be summarized as the following:

(23) *Dropping Criterion*: If $B_{0\cdot}^{-1} A_{\cdot s}^* < 0$ for some column index s not in the initial basis matrix and $B_{m\cdot}^{-1} A_{\cdot s}^* > 0$ for at least one $m \geq 1$, then add column s to the basis matrix and drop any column r such that column r attains the minimum in (22); i.e., r is such that $r \geq 1$ and

$$(24) \quad B_{r\cdot}^{-1} b^* / B_{r\cdot}^{-1} A_{\cdot s}^* \\ = \min_m \{B_{m\cdot}^{-1} b^* / B_{m\cdot}^{-1} A_{\cdot s}^* : m \geq 1 \text{ and } m \text{ is such that } B_{m\cdot}^{-1} A_{\cdot s}^* > 0\}.$$

Using the nondegeneracy assumption, it can be seen that x_s^* defined by (22) is positive and introducing column s into the basis matrix leads to a strict increase in the objective function for the LP (10).

If the minimum in (22) or (24) is attained by a unique column r , then it can be seen that the new basis matrix is uniquely determined. It is also possible to show that the new basis matrix will also satisfy the nondegeneracy assumption.

The above criteria form the core of an effective algorithm for solving the LP (10).⁴ Obviously, a finite but tremendously inefficient algorithm for solving (10) would involve solving the following system of M equations in M unknowns:

⁴ There are a few gaps but we will fill them in later.

$$(25) \sum_{j=1}^M A_{n_j} x_{n_j} = b$$

for all possible choices of M of the N columns of the A matrix, checking to see if the resulting x_{n_j} are nonnegative, evaluating the objective function at these basic feasible solutions and then the nonnegative solution that gave the lowest objective function $\sum_{j=1}^M c_{n_j} x_{n_j}$ would be picked. What the simplex algorithm does is to search through basis matrices in such a way that we always increase our objective function.⁵ Experience with the simplex algorithm has shown that an optimal basis matrix is generally reached in the order of M to $3M$ iterations of the algorithm. *Thus the simplex algorithm is tremendously more efficient than simply searching through all of the basic feasible solutions for the linear program.*

4. An Example of the Simplex Algorithm

Let us use the simplex algorithm to solve the LP that was graphed in section 2 above. Rewriting the problem in the form (10) leads to the following problem:

$$(26) \max_{x_0, x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, s_2 \geq 0} \{x_0 : e_0 x_0 + A_{\bullet 1}^* x_1 + A_{\bullet 2}^* x_2 + A_{\bullet 3}^* s_1 + A_{\bullet 4}^* s_2 = b^*\}$$

$$\text{where } e_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, A_{\bullet 1}^* = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, A_{\bullet 2}^* = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, A_{\bullet 3}^* = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, A_{\bullet 4}^* = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \text{ and } b^* = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Define the initial basis matrix to be

$$(27) B \equiv [e_0, A_{\bullet 1}^*, A_{\bullet 2}^*] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \equiv \begin{bmatrix} 1 & c^T \\ 0_2 & A \end{bmatrix}$$

where A is a two by two matrix. Using determinants, if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, we know that the formula for A^{-1} is given by (provided that $a_{11}a_{22} - a_{12}a_{21} \neq 0$):

$$(28) A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ for our example.}$$

Using (28), we can readily calculate B^{-1} as follows:

⁵ In general, we either increase the objective function or leave it unchanged at each iteration of the simplex algorithm or we find an unbounded solution.

$$(29) B^{-1} = \begin{bmatrix} 1 & c^T \\ 0_2 & A \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -c^T A^{-1} \\ 0_2 & A^{-1} \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check for feasibility:

$$(30) \begin{bmatrix} x_0^* \\ x_1^* \\ x_2^* \end{bmatrix} = B^{-1}b^* = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ 3 \end{bmatrix}.$$

To check that we have a basic feasible solution that satisfies the nonnegativity constraints, we need to check that $x_1^* \geq 0$ and $x_2^* \geq 0$. Equations (30) show that these constraints are indeed satisfied.

Check for optimality:

We need to check that the inner product of the first row of B^{-1} with each column not in the basis matrix is positive or zero.

$$(31) B_0^{-1}A_{.3}^* = [1, -2, -1] \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = 2 > 0; \quad B_0^{-1}A_{.4}^* = [1, -2, -1] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = 1 > 0.$$

The optimality conditions are satisfied and thus $x_1^* = 1$ and $x_2^* = 3$ solve the LP.

In the above example, it was easy to find a starting basic feasible solution so that we could start the simplex algorithm. However, finding a starting basic feasible solution may not be all that easy in a complicated problem. In fact, it may be the case that a given LP may not even have a feasible solution. Thus in the following section, we discuss Dantzig's Phase I procedure for finding a starting basic feasible solution.

5. Finding a Starting Basic Feasible Solution

Consider again the first form (1) of a linear programming problem, $\min_x \{c^T x : Ax = b; x \geq 0_N\}$. Now multiply both sides of each constraint equation through by -1 if necessary so that the right hand side vector b is nonnegative; i.e., so that $b \geq 0_M$.

Now consider the following *Phase I* LP where we have introduced a vector of *artificial variables* s :

$$(32) \min_{x,s} \{1_M^T s : Ax + I_M s = b; x \geq 0_N; s \geq 0_M\}$$

where 1_M is an M dimensional column vector of ones and I_M is an M by M identity matrix. A basic feasible starting solution to (32) is $s = b \geq 0_M$ and $x = 0_N$. Note that this starting basis matrix is nonsingular. Now use the simplex algorithm to solve (32).⁶

Several cases can occur.

Case 1: The minimum for (32) is greater than 0.

In this case, there is no feasible solution for the original LP and we can stop at this point.

Case 2: The minimum for (32) equals 0.

In this case, the original LP has feasible solutions. However, we need to consider two subcases.

Case 2A: The minimum for (32) is 0 and no columns corresponding to the artificial variables are in the final basis matrix.

In this case, we have an $x^0 \geq 0_N$ such that $Ax^0 = b$ and at most M components of x^0 are nonzero. Thus we have a starting basic feasible solution for the original LP (1).

Case 2B: The minimum for (32) is 0 and one or more columns corresponding to the artificial variables are in the final basis matrix.

In this case, we have a starting basic feasible solution for the following linear programming problem, which turns out to be equivalent to the original problem (1):

$$(33) \max_{x_0, x \geq 0_N, s \geq 0_M} \{x_0 : x_0 + c^T x = 0 ; Ax + I_M s = b ; 1_M^T s = 0\}.$$

Note that (33) is the same as (1) except that the vector of artificial variables s has been inserted into the constraint equations $Ax = b$ and the extra constraint $1_M^T s = 0$ has been added to the problem, along with the nonnegativity restrictions on s , $s \geq 0_M$. However, note that the constraints $1_M^T s = 0$ and $s \geq 0_M$ together imply that $s = 0_M$. Thus the artificial variables will be kept at 0 levels for all iterations of the simplex algorithm applied to (33) and so solving (33) will also solve (1).

Thus solving the Phase I linear programming problem (32) will tell us whether a feasible solution to the original problem (1) exists or not and if a feasible solution to the original problem does exist, the solution to the Phase I problem will give us a starting basic feasible solution to solve the original linear programming problem (1), which is called the *Phase II problem* by Dantzig (1963).

⁶ Note that the unbounded solution case cannot occur for this Phase I problem because the objective function is bounded from below by 0.

We conclude this section by solving the Phase I problem for the example in the previous section. Our Phase I problem is the following one:

$$(34) \max_{x_0, x_1 \geq 0, x_2 \geq 0, s_1 \geq 0, s_2 \geq 0} \{x_0 : \\ e_0 x_0 + A_{.1}^* x_1 + A_{.2}^* x_2 + A_{.3}^* x_3 + A_{.4}^* x_4 + A_{.5}^* s_1 + A_{.6}^* s_2 = b^*\}$$

$$\text{where } e_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, A_{.1}^* = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, A_{.2}^* = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, A_{.3}^* = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, A_{.4}^* = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, A_{.5}^* = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \\ A_{.6}^* = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ and } b^* = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Our starting basis matrix is B defined as follows:

$$(35) B = [e_0, A_{.5}^*, A_{.6}^*] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and so } B^{-1} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Feasibility Check:

$$(36) \begin{bmatrix} x_0^{(0)} \\ s_1^{(0)} \\ s_2^{(0)} \end{bmatrix} = B^{-1} b^* = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}.$$

Since $s_1^{(0)} \geq 0$ and $s_2^{(0)} \geq 0$, the feasibility conditions are satisfied.

Optimality Check:

$$(37) \begin{aligned} B_{0.}^{-1} A_{.1}^* &= 0 ; \text{ OK} \\ B_{0.}^{-1} A_{.2}^* &= -1 ; \text{ Can introduce } A_{.2}^* \text{ into the basis matrix} \\ B_{0.}^{-1} A_{.3}^* &= 1 ; \text{ OK} \\ B_{0.}^{-1} A_{.4}^* &= 1 ; \text{ OK} \\ B_{0.}^{-1} A_{.5}^* &= 0 ; \text{ OK (must equal 0 since } A_{.5}^* \text{ is in the initial basis matrix)} \\ B_{0.}^{-1} A_{.6}^* &= 0 ; \text{ OK (must equal 0 since } A_{.6}^* \text{ is in the initial basis matrix).} \end{aligned}$$

There is only one column $A_{.s}^*$ such that $B_{0.}^{-1} A_{.s}^* < 0$, namely $A_{.2}^*$. If there were two or more columns with $B_{0.}^{-1} A_{.s}^* < 0$, then pick the s such that $B_{0.}^{-1} A_{.s}^*$ is the most negative. Thus column $A_{.2}^*$ will enter the new basis matrix but which column will leave? We need to calculate:

Dropping Criterion:

$$\begin{aligned}
 (38) \ x_2^{(1)} &\equiv \min_m \{B_{m \cdot}^{-1} b^* / B_{m \cdot}^{-1} A_{\cdot 2}^* : m \geq 1 \text{ and } m \text{ is such that } B_{m \cdot}^{-1} A_{\cdot 2}^* > 0\} \\
 &= B_{2 \cdot}^{-1} b^* / B_{2 \cdot}^{-1} A_{\cdot 2}^* \\
 &= 2/1 \\
 &= 2
 \end{aligned}$$

since $B_{1 \cdot}^{-1} A_{\cdot 2}^* = 0$, $B_{2 \cdot}^{-1} A_{\cdot 2}^* = 1$ and $B_{2 \cdot}^{-1} b^* = 2$. In other words, the minimum in (38) is attained for $m = 2$ and the corresponding column of the original B matrix leaves the basis matrix; i.e., the last column $A_{\cdot 6}^*$ leaves. Thus the new basis matrix becomes:

$$(39) \ B^{(1)} \equiv [e_0, A_{\cdot 5}^*, A_{\cdot 2}^*] = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and so } [B^{(1)}]^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Feasibility Check:

$$(40) \ \begin{bmatrix} x_0^{(1)} \\ s_1^{(1)} \\ x_2^{(1)} \end{bmatrix} = B^{(1)-1} b^* = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

Since $s_1^{(1)} \geq 0$ and $x_2^{(1)} \geq 0$, the feasibility conditions are satisfied.

Optimality Check:

$$\begin{aligned}
 (41) \ B_{0 \cdot}^{(1)-1} A_{\cdot 1}^* &= -1 ; \text{ Can introduce } A_{\cdot 1}^* \text{ into the basis matrix} \\
 B_{0 \cdot}^{(1)-1} A_{\cdot 2}^* &= 0 ; \text{ OK (must equal 0 since } A_{\cdot 2}^* \text{ is in the basis matrix)} \\
 B_{0 \cdot}^{(1)-1} A_{\cdot 3}^* &= 1 ; \text{ OK} \\
 B_{0 \cdot}^{(1)-1} A_{\cdot 4}^* &= 0 ; \text{ OK} \\
 B_{0 \cdot}^{(1)-1} A_{\cdot 5}^* &= 0 ; \text{ OK (must equal 0 since } A_{\cdot 5}^* \text{ is in the basis matrix)} \\
 B_{0 \cdot}^{(1)-1} A_{\cdot 6}^* &= 1 ; \text{ OK.}
 \end{aligned}$$

There is only one column $A_{\cdot s}^*$ such that $B_{0 \cdot}^{(1)-1} A_{\cdot s}^* < 0$, namely $A_{\cdot 1}^*$. Thus column $A_{\cdot 1}^*$ will enter the new basis matrix but which column will leave? We need to calculate:

Dropping Criterion:

$$\begin{aligned}
 (42) \ x_1^{(2)} &\equiv \min_m \{B_{m \cdot}^{(1)-1} b^* / B_{m \cdot}^{(1)-1} A_{\cdot 1}^* : m \geq 1 \text{ and } m \text{ is such that } B_{m \cdot}^{-1} A_{\cdot 1}^* > 0\} \\
 &= B_{1 \cdot}^{(1)-1} b^* / B_{1 \cdot}^{(1)-1} A_{\cdot 1}^* \\
 &= 1/1 \\
 &= 1
 \end{aligned}$$

since $B_{1\cdot}^{(1)-1}A_{\cdot 1}^* = 1$, $B_{2\cdot}^{(1)-1}A_{\cdot 1}^* = -1$ and $B_{1\cdot}^{(1)-1}b^* = 1$. In other words, the minimum in (42) is attained for $m = 1$ and the corresponding column of the $B^{(1)}$ matrix leaves the basis matrix; i.e., the second column $A_{\cdot 5}^*$ leaves and $A_{\cdot 1}^*$ enters. Thus the new basis matrix becomes:

$$(43) B^{(2)} \equiv [e_0, A_{\cdot 1}^*, A_{\cdot 2}^*] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \text{ and so } [B^{(2)}]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Feasibility Check:

$$(44) \begin{bmatrix} x_0^{(2)} \\ x_1^{(2)} \\ x_2^{(2)} \end{bmatrix} = B^{(2)-1}b^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$$

Since $x_1^{(2)} \geq 0$ and $x_2^{(2)} \geq 0$, the feasibility conditions are satisfied.

Optimality Check:

$$(45) \begin{aligned} B_{0\cdot}^{(2)-1}A_{\cdot 1}^* &= 0 ; \text{ OK (must equal 0 since } A_{\cdot 2}^* \text{ is in the basis matrix)} \\ B_{0\cdot}^{(2)-1}A_{\cdot 2}^* &= 0 ; \text{ OK (must equal 0 since } A_{\cdot 2}^* \text{ is in the basis matrix)} \\ B_{0\cdot}^{(2)-1}A_{\cdot 3}^* &= 0 ; \text{ OK} \\ B_{0\cdot}^{(2)-1}A_{\cdot 4}^* &= 0 ; \text{ OK} \\ B_{0\cdot}^{(2)-1}A_{\cdot 5}^* &= 1 ; \text{ OK} \\ B_{0\cdot}^{(2)-1}A_{\cdot 6}^* &= 1 ; \text{ OK.} \end{aligned}$$

Thus the optimality conditions for solving the Phase I problem are satisfied with $A_{\cdot 1}^*$ and $A_{\cdot 2}^*$ the final columns in the optimal basis matrix. Note that the objective function x_0 increased at each iteration of the Phase I problem and we ended up with a basis matrix that we could use to start the Phase II problem. Note also that the basis matrix was invertible at each stage of the Phase I algorithm. In the following section, we will show that the basis matrix is always invertible, provided that we start with a basis matrix that is invertible, which we can always do. Thus our nonsingular basis matrix assumption made in section 2 is not a restrictive assumption.

6. Nonsingularity of the Basis Matrix in the Simplex Algorithm

Suppose that at some iteration of the simplex algorithm, the basis matrix is $B \equiv [e_0, A_{\cdot 1}^*, \dots, A_{\cdot M}^*]$ is nonsingular; i.e., B^{-1} exists. If $B_{0\cdot}^{-1}A_{\cdot n}^* \geq 0$ for $n = 1, 2, \dots, N$, then we have an optimal solution and the algorithm stops. If $B_{0\cdot}^{-1}A_{\cdot s}^* < 0$ for some s with $B_{m\cdot}^{-1}A_{\cdot s}^* \leq 0$ for $m = 1, 2, \dots, M$, then we have an unbounded optimal solution and the algorithm stops. Now consider the remaining case; i.e., suppose that there exists a column index s such that

$$(46) B_{\bullet 0}^{-1} A_{\bullet s}^* < 0 \text{ and } B_{\bullet m}^{-1} A_{\bullet s}^* > 0 \text{ for at least one } m \geq 1.$$

Thus $A_{\bullet s}^*$ enters the basis matrix and according to the dropping criterion, $A_{\bullet r}^*$ leaves the basis matrix where r is such that $B_{r\bullet}^{-1} b^* / B_{r\bullet}^{-1} A_{\bullet s}^* = \min_m \{B_{m\bullet}^{-1} b^* / B_{m\bullet}^{-1} A_{\bullet s}^* : m \geq 1 \text{ and } m \text{ is such that } B_{m\bullet}^{-1} A_{\bullet s}^* > 0\}$. Thus we have:

$$(47) B_{r\bullet}^{-1} A_{\bullet s}^* > 0.$$

Since the initial basis matrix $B \equiv [e_0, A_{\bullet 1}^*, \dots, A_{\bullet M}^*]$ is nonsingular, it must be possible to express the entering column $A_{\bullet s}^*$ as a linear combination of the columns of B ; i.e., there exists $\alpha \equiv [\alpha_0, \alpha_1, \dots, \alpha_M]^T$ such that

$$(48) A_{\bullet s}^* = B\alpha \text{ or } \alpha = B^{-1} A_{\bullet s}^*.$$

Using (47) and (48), we have

$$(49) \alpha_r = B_{r\bullet}^{-1} A_{\bullet s}^* > 0.$$

Now form the new basis matrix $B^{(1)}$ defined as follows:

$$(50) B^{(1)} \equiv [e_0, A_{\bullet 1}^*, A_{\bullet 2}^*, \dots, A_{\bullet r-1}^*, A_{\bullet s}^*, A_{\bullet r+1}^*, A_{\bullet r+2}^*, \dots, A_{\bullet M}^*].$$

It is easy to see that (48) and $\alpha_r > 0$ imply that $A_{\bullet r}^*$ can be expressed as a linear combination of the columns of $B^{(1)}$:

$$(51) A_{\bullet r}^* = B^{(1)}\beta; \quad \beta \equiv [\beta_0, \beta_1, \dots, \beta_M]^T$$

where

$$(52) \beta_r = 1/\alpha_r = 1/B_{r\bullet}^{-1} A_{\bullet s}^* \text{ and } \beta_m = -\alpha_m/\alpha_r = -B_{m\bullet}^{-1} A_{\bullet s}^* / B_{r\bullet}^{-1} A_{\bullet s}^* \text{ for } m \neq r.$$

Comparing B with $B^{(1)}$, it can be seen that

$$\begin{aligned} (53) B &\equiv [e_0, A_{\bullet 1}^*, \dots, A_{\bullet r-1}^*, A_{\bullet r}^*, A_{\bullet r+1}^*, \dots, A_{\bullet M}^*] \\ &= [e_0, A_{\bullet 1}^*, \dots, A_{\bullet r-1}^*, A_{\bullet s}^*, A_{\bullet r+1}^*, \dots, A_{\bullet M}^*][e_0, e_1, \dots, e_{r-1}, \beta, e_{r+1}, \dots, e_M] \\ &= B^{(1)}E \end{aligned}$$

where $E \equiv [e_0, e_1, \dots, e_{r-1}, \beta, e_{r+1}, \dots, e_M]$ and the components of β are defined by (52). Note that since $\beta_r = 1/\alpha_r > 0$, the determinant of E is equal to $\beta_r > 0$ and hence E^{-1} exists. Thus $B^{(1)} = BE^{-1}$ and

$$(54) B^{(1)-1} = EB^{-1}$$

and hence the inverse of the new basis matrix exists. Thus we have proven:

Theorem 2; Dantzig (1963): The basis matrix is nonsingular at each iteration of the simplex algorithm.

Note that the inverse of the new basis matrix $B^{(1)}$ is easy to compute using (54): we need only premultiply the inverse of the old basis matrix B^{-1} by the matrix $E \equiv [e_0, e_1, \dots, e_{r-1}, \beta, e_{r+1}, \dots, e_M]$, which differs from the identity matrix in only one column.⁷

There is one remaining detail to be filled in to complete our discussion of Dantzig's simplex algorithm for solving a linear program.

7. The Degeneracy Problem

Recall the nondegeneracy assumption made in section 3. Recall that degeneracy occurs when the dropping criterion does not yield a *unique* column to be dropped; i.e., when we compute the minimum in (24), the minimum is attained by more than one column index m . When this situation occurs, then it can happen at the next stage of the simplex algorithm that we do not get a drop in the objective function and two or more columns cycle in and out of the basis matrix without causing a drop in the objective function. Thus there is the theoretical possibility that the simplex algorithm could enter an infinite cycle and the algorithm might fail to converge.

It seems from an a priori point of view that it would be extremely unlikely that this cycling phenomenon could occur. However, Dantzig reports that the degeneracy phenomenon is common:

“It is common experience, based on the solutions of thousands of practical linear programming problems by the simplex method, that nearly every problem at some stage of the process is degenerate.” George Dantzig (1963; 231).

When degeneracy occurs, it is conceivable that we could have a sequence of basis matrices where the objective function does not change such that the initial basis matrix reappears at some stage, which could then lead to an endless cycle of the same sequence of basis matrices. There are at least two artificially constructed examples of linear programming problems where this cycling phenomenon occurred; see Hoffman (1953) and Beale (1955). Dantzig (1963; 228-230) presents these two examples but he comments on the phenomenon as follows:

“To date, there has not been one single case of circling, except in the specially constructed examples of Hoffman and Beale. Apparently, circling is a very rare phenomenon in practice. For this reason, most instruction codes for electronic computers use no special device for perturbing the problem to avoid degeneracy and the possibility of circling.” George Dantzig (1963; 231).

⁷ See (52) for the definition of the components of the β vector.

It turns out that cycling or circling can be avoided if we try dropping a different sequence of columns after going through a cycle of column changes where the objective function did not change the first time around. Dantzig (1963) develops various rules for choosing which column to drop which work and he has references to rules developed by others. For our purposes, the exact form of the dropping rule is not important; all we need to know is that a dropping rule exists such that the simplex algorithm will terminate in a finite number of iterations, even when nondegeneracy is not assumed.

We turn now to a topic of great economic interest.

8. The Dual Linear Program

In this section, we use the third formulation for a linear programming problem as our starting point. Thus we suppose that we are given as data an M by N matrix A , an N dimensional vector c and an M dimensional vector b . Then the *primal linear programming problem* is defined as the problem of maximizing the linear function $c^T x$ with respect to the vector of primal decision variables x subject to the inequality constraints $Ax \leq b$ and the nonnegativity constraints $x \geq 0_N$:

$$(55) \text{ Primal Problem: } \max_x \{c^T x : Ax \leq b ; x \geq 0_N\}.$$

The *dual* to the above problem switches the roles of b and c , changes a maximization problem into a minimization problem, switches from a vector of primal variables x that operate on the columns of A to a vector of dual variables y which operate on the rows of A and writes the inequality constraints in the opposite direction:

$$(56) \text{ Dual Problem: } \min_y \{y^T b : y^T A \geq c^T ; y \geq 0_M\}.$$

Note that the dual problem is also a linear programming problem. The primal and dual problems are related by the following theorem:

Theorem 3; von Neumann (1947), Dantzig (1963; chapter 6): If the primal problem has a finite optimal solution x^* say, then the dual problem also has a finite optimal solution y^* say and moreover:

$$(57) c^T x^* = y^{*T} b ;$$

i.e., the optimal values of the primal and dual objective functions are equal.

Proof: We first put the primal problem into the standard simplex algorithm format. Define $x_0 \equiv c^T x$. After adding the vector of slack variables s to Ax , we find that the primal problem (55) is equivalent to the following problem:

$$(58) \max_{x_0, x \geq 0_N, s \geq 0_M} \{x_0 : \begin{bmatrix} 1 \\ 0_M \end{bmatrix} x_0 + \begin{bmatrix} -c^T & 0_M^T \\ A & I_M \end{bmatrix} \begin{bmatrix} x \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix} \}.$$

Apply Phase I of the simplex algorithm to the LP defined by (58). Since we assume that the primal problem has a finite optimal solution, the optimal objective function for the Phase I problem will attain its lower bound of 0. Thus we will have:

$$(59) \quad 0 = \max_{x_0, x \geq 0_N, s \geq 0_M, z \geq 0_M} \left\{ x_0 : \begin{bmatrix} 1 \\ 0_M \end{bmatrix} x_0 + \begin{bmatrix} 0_N^T & 0_M^T & 1_M^T \\ A & I_M & -I_M \end{bmatrix} \begin{bmatrix} x \\ s \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix} \right\}$$

where we have added the vector of artificial variables $z \geq 0_M$ to the constraints of (58). There are two cases that can occur at this point.

Case I: In this case, the Phase I problem yields a nonsingular initial basis matrix for the Phase II problem (58), which does not have any columns in it that correspond to the artificial variables z . Thus in this case, we may apply the simplex algorithm to (58) without including any artificial variables and since by assumption, there is a finite optimal solution to (58), the simplex algorithm will terminate and there will exist an $M+1$ by $M+1$ nonsingular basis matrix B (where the first column of B is the unit vector e_0) such that the following *optimality conditions* hold:

$$(60) \quad B_0^{-1} \begin{bmatrix} -c^T & 0_M^T \\ A & I_M \end{bmatrix} \geq [0_N^T, 0_M^T] \quad \text{and} \quad B_0^{-1} e_0 = 1$$

where $B_0^{-1} = [1, y^{*T}]$ say, is the first row of B^{-1} .⁸ Thus the optimality conditions (60) imply that y^* satisfies the following inequalities:

$$(61) \quad y^{*T} A \geq c^T \quad \text{and} \quad y^{*T} I_M \geq 0_M^T.$$

Thus y^* is a feasible solution for the dual problem (56).

At this point, we need the following *auxiliary result*: if x^* and y^* are feasible solutions for the primal and dual problems respectively, then we have the following bounds for the objective functions for the two problems:

$$(62) \quad c^T x^* \leq y^{*T} b;$$

i.e., the dual objective function evaluated at a feasible solution to the dual, $y^{*T} b$, is an upper bound for the primal objective function evaluated at any feasible solution and the primal objective function evaluated at a feasible solution for the primal, $c^T x^*$, is a lower bound for the dual objective function evaluated at any feasible solution. To prove (62), note that x^* feasible for the primal problem and y^* feasible for the dual problem means that x^* and y^* satisfy the following inequalities:

⁸ We can deduce that the first component of B_0^{-1} is 1 using $B_0^{-1} e_0 = 1$ since e_0 is the first column of B .

$$(63) \ x^* \geq 0_N ; Ax^* \leq b ; y^* \geq 0_M ; y^{*T}A \geq c^T .$$

Thus

$$\begin{aligned} (64) \ c^T x^* &\leq c^T x^* + y^{*T}(b - Ax^*) && \text{since } y^* \geq 0_M \text{ and } Ax^* \leq b \\ &= y^{*T}b + (c^T - y^{*T}A)x^* && \text{rearranging terms} \\ &\leq y^{*T}b && \text{since } x^* \geq 0_N \text{ and } y^{*T}A \geq c^T . \end{aligned}$$

Now (61) shows that the y^* defined by $B_0^{-1} \equiv [1, y^{*T}]$ is a feasible solution for the dual. In view of (64), we need only show that $y^{*T}b = c^T x^*$ where x^* solves the primal, and we will have y^* as an optimal solution to the dual problem.

Substitute the optimal Phase II solution to (58), x_0^* , x^* and s^* into the constraints listed in (58) and then premultiply both sides of this matrix equation by $B_0^{-1} \equiv [1, y^{*T}]$. We obtain the following equation:

$$(65) \ [1, y^{*T}] \begin{bmatrix} 1 \\ 0_M \end{bmatrix} x_0^* + [1, y^{*T}] \begin{bmatrix} -c^T & 0_M^T \\ A & I_M \end{bmatrix} \begin{bmatrix} x^* \\ s^* \end{bmatrix} = [1, y^{*T}] \begin{bmatrix} 0 \\ b \end{bmatrix} .$$

Obviously, $[1, y^{*T}][1, 0_M^T]^T = 1$ and so the first term on the left hand side of (65) is simply equal to x_0^* . It turns out that the next set of terms on the left hand side of (65) is equal to 0 since if a particular column in the matrix $\begin{bmatrix} -c^T & 0_M^T \\ A & I_M \end{bmatrix}$ is in the optimal basis matrix, the inner product of this column with the first row of B^{-1} , $[1, y^{*T}]$, will be 0⁹ and if a particular column in this matrix is *not* in the optimal basis matrix, then the corresponding x_n^* or s_m^* is zero and again the term will be 0. The term on the right hand side of (65) is equal to $y^{*T}b$. Thus (65) simplifies to:

$$(66) \ x_0^* = y^{*T}b .$$

Since $x_0^* = c^T x^*$, we have $c^T x^* = y^{*T}b$ and thus y^* is indeed an optimal solution to the dual problem.

Case 2: In this case, the solution to the Phase I problem contains one or more columns corresponding to the artificial variables z . In this case, the Phase I solution gives us a starting basic feasible solution to the following problem, which is equivalent to the primal problem (55):

⁹ This follows from $B^{-1}B = I_{M+1}$.

$$(67) \max_{x_0, x \geq 0_N, s \geq 0_M, z \geq 0_M} \left\{ x_0 : \begin{bmatrix} 1 \\ 0_M \\ 0 \end{bmatrix} x_0 + \begin{bmatrix} -c^T \\ A \\ 0_N^T \end{bmatrix} x + \begin{bmatrix} 0_M^T \\ I_M \\ 0_M^T \end{bmatrix} s + \begin{bmatrix} 0_M^T \\ -I_M \\ 1_M^T \end{bmatrix} z = \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} \right\}.$$

We have set up the artificial variables in a slightly different way than was suggested in section 5 above. Since we already have added $+ I_M s$ to Ax , we added $- I_M z$ as the artificial variables. Between the s and z variables, we can obviously find a Phase I starting basic feasible solution to the constraints $Ax + I_M s - I_M z = b$, no matter what the signs are for the components of the b vector.

Now apply the Phase II simplex algorithm to (67). The constraints $1_M^T z = 0$ and $z \geq 0_M$ will force the z variables to be kept at 0 levels at all iterations of the Phase II algorithm. Since we assumed that a finite optimal solution to the original primal problem existed, the Phase II simplex algorithm will eventually terminate and there will exist an $M+2$ by $M+2$ basis matrix C such that the following optimality conditions will hold:

$$(68) C_0 \cdot^{-1} \begin{bmatrix} -c^T & 0_M^T & 0_M^T \\ A & I_M & -I_M \\ 0_N^T & 0_M^T & 1_M^T \end{bmatrix} \geq [0_N^T, 0_M^T, 0_M^T]$$

where $C_0 \cdot^{-1} \equiv [1, y^{*T}, y_0^*]$ is the first row of C^{-1} . The optimal solution to (67), x_0^*, x^*, s^*, z^* will satisfy the following relations:

$$(69) x_0^* = c^T x^*; x^* \geq 0_N; s^* \geq 0_M; z^* = 0_M.$$

The first two sets of inequalities in (68) imply that y^* satisfies:

$$(70) y^{*T} A \geq c^T \text{ and } y^* \geq 0_M$$

which implies that y^* is feasible for the dual problem. Now evaluate the constraints in (67) at the optimal Phase II solution, x_0^*, x^*, s^*, z^* and premultiply the constraint equations through by $C_0 \cdot^{-1} \equiv [1, y^{*T}, y_0^*]$. We find that as in Case 1:

$$(71) x_0^* + 0 = y^{*T} b$$

Where the second term is 0 because columns not in the final Phase II basis matrix are multiplied by 0 while the columns in the final basis matrix have a 0 inner product with the first row of the inverse of the basis matrix, $C_0 \cdot^{-1} \equiv [1, y^{*T}, y_0^*]$. Since x^* is feasible (and optimal) for the primal problem and y^* is feasible for the dual problem, (69) and (71) and the auxiliary result imply that y^* is a solution to the dual with $y^{*T} b = c^T x^*$. Q.E.D.

Corollary: If the primal problem has a feasible solution and the dual problem has a feasible solution, then both the primal and dual problems have finite optimal solutions and the values of the optimal objective functions coincide.

Proof: The auxiliary result (62) shows that if the primal and dual both have feasible solutions, then the primal objective function is bounded from above. Hence the unbounded solution case for the primal cannot occur. Now repeat the proof of Theorem 3. Q.E.D.

Note that the above proof of the duality theorem for linear programs is entirely algebraic and rests on the mechanics of the simplex algorithm for solving linear programs. The proof of the above theorem is essentially due to Dantzig (1963).

A further implication of the above duality theorem is that if the primal solution has an unbounded optimal solution, then the dual cannot have a feasible solution.

The duality theorem for linear programs has many economic applications as will be seen by studying the problems at the end of the chapter.

We conclude this section by proving another result that will be helpful in providing an economic interpretation for the dual variables.

Theorem 4: Basis Theorem for Linear Programs; Dantzig (1963; 121): Suppose that a nonsingular basis matrix B is *optimal* for the problem:

$$(72) \max_{x_0, x} \{x_0 : e_0 x_0 + A^* x = b^* ; x \geq 0_N\}.$$

Suppose further that this basis matrix B generates a *feasible* solution for the following problem where the right hand side vector b^* has been replaced by b^1 :

$$(73) \max_{x_0, x} \{x_0 : e_0 x_0 + A^* x = b^1 ; x \geq 0_N\}.$$

Then the basis matrix B is also optimal for the new problem (73).

Proof: Since B is an optimal basis matrix for (72), then we have:

$$(74) B_m^{-1} b^* \geq 0 \text{ for } m = 1, 2, \dots, M \text{ (feasibility conditions satisfied);}$$

$$(75) B_0^{-1} A_n^* \geq 0 \text{ for } n = 1, 2, \dots, N \text{ (optimality conditions satisfied).}$$

But by assumption, the initial optimal basis matrix B generates a feasible solution for the new b vector, b^1 , so that we have:

$$(76) B_m^{-1} b^1 \geq 0 \text{ for } m = 1, 2, \dots, M.$$

But since the basis matrix has not changed, the optimality conditions (75) are still satisfied (these conditions do not depend directly on b) and so (75) and (76) imply that the old basis matrix B is still optimal for the new problem (73). Q.E.D.

Theorem 4 is helpful in providing an economic interpretation for the dual variables. However, to see the connection, we have to consider a slightly different primal and dual LP problems compared to (55) and (56). Thus consider an equality constrained primal problem of the following form:

$$(77) \max_x \{ c^T x : Ax = b ; x \geq 0_N \}.$$

In order to find the dual problem to (77), we need to write the equality constraints $Ax = b$ in an inequality format. Thus (77) is equivalent to the following problem:

$$(78) \max_x \{ c^T x : \begin{bmatrix} A \\ -A \end{bmatrix} x \leq \begin{bmatrix} b \\ -b \end{bmatrix} ; x \geq 0_N \}.$$

Using our rules for forming the dual problem, it can be seen that the dual to (78) is:

$$(79) \min_{y^1, y^2} \{ y^{1T} b - y^{2T} b : [y^{1T}, y^{2T}] \begin{bmatrix} A \\ -A \end{bmatrix} \geq c^T ; y^1 \geq 0_M ; y^2 \geq 0_M \}.$$

Define the unrestricted vector of variables $y \equiv y^1 - y^2$ and using this definition, we can rewrite the dual problem (79) as follows:

$$(80) \min_y \{ y^T b : y^T A \geq c^T \}.$$

Thus (80) is the dual problem to the primal problem (77): *equality constraints in the primal lead to a dual problem with unrestricted in sign dual variables* (instead of nonnegative dual variables as before).

Assume that there is a finite optimal solution to (77). Putting (77) into the standard simplex algorithm format leads to the following equivalent problem:

$$(81) \max_{x_0, x} \{ x_0 : \begin{bmatrix} 1 \\ 0_M \end{bmatrix} x_0 + \begin{bmatrix} -c^T \\ A \end{bmatrix} x = \begin{bmatrix} 0 \\ b \end{bmatrix} ; x \geq 0_N \}.$$

Suppose that the final optimal basis matrix for (81) turns out to include the first M columns of the A matrix. Thus we have

$$(82) B = \begin{bmatrix} 1 & -c_1 & \dots & -c_M \\ 0_M & A_{\bullet 1} & \dots & A_{\bullet M} \end{bmatrix}$$

and we assume that B is nonsingular. Our final assumption is that the final basis matrix B for (81) is *nondegenerate* so that

$$(83) \ x_m^* = B_{m\bullet}^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} > 0 \text{ for } m = 1, 2, \dots, M.$$

We may repeat the proof of the Duality Theorem for linear programs in this set up. As before, we find that

$$(84) \ B_{0\bullet}^{-1} = [1, y^{*T}]$$

where y^* turns out to be a solution to the dual problem (80). The nondegeneracy assumptions (83) imply that y^* must satisfy the following equations:

$$(85) \ [1, y^{*T}] \begin{bmatrix} -c_m \\ A_{\bullet m} \end{bmatrix} = 0 \text{ for } m = 1, 2, \dots, M.$$

Equations (85) imply that the dual variables must be unique. Thus *nondegeneracy of the final basis matrix implies that the optimal solution to the dual problem is unique*.

Recall that the feasibility restrictions (83) are satisfied by our basis matrix, which we rewrite as follows:

$$(86) \ B_{m\bullet}^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} > 0 \text{ for } m = 1, 2, \dots, M.$$

The following optimality conditions are also satisfied by our basis matrix B :

$$(87) \ B_{0\bullet}^{-1} \begin{bmatrix} -c_n \\ A_{\bullet n} \end{bmatrix} \geq 0 \text{ for } n = 1, 2, \dots, N.$$

Now we can apply the Basis Theorem to the above setup. If b^1 is sufficient close to the initial b , the nonnegativity restrictions (86) will continue to hold using the old basis matrix B ; i.e., we will have, b^1 close to b :

$$(88) \ B_{m\bullet}^{-1} \begin{bmatrix} 0 \\ b^1 \end{bmatrix} > 0 \text{ for } m = 1, 2, \dots, M.$$

The optimality conditions (87) will continue to hold for this new primal problem where b^1 replaces b and hence the basis matrix B continues to be optimal for the new problem. Setting $x_{M+1} = x_{M+2} = \dots = x_N = 0$ for the constraints in (81) and premultiplying both sides of the constraints by $[1, y^{*T}]$ where y^* is defined by (84) leads to the following equation, where we have also used (85):

$$(89) x_0 = y^{*T}b.$$

All of this shows that for b^1 close to b , the optimal primal and dual objective functions, regarded as functions of b^1 , are given by

$$(90) V(b^1) = y^{*T}b^1.$$

Thus for $b^1 = b$, the vector of first order partial derivatives of the optimized primal and dual objective functions, with respect to the components of b is equal to:

$$(91) \nabla_b V(b) = y^*$$

where the unique vector of dual variables y^* is defined by (84). Thus y_m^* is the marginal increase in the primal (and dual) objective functions due to a small increase in the m th component of b , b_m , provided that we have a nondegenerate primal solution. Note that this interpretation for a dual variable is entirely analogous to the economic interpretation of a Lagrange multiplier in classical constrained optimization.¹⁰

9. The Geometric Interpretation of a Linear Program in Requirements Space

Recall the basic linear program defined by (2) above and recall that we rewrote in the form (10) above. For convenience, we repeat (10) below as problem (92):

$$(92) \max_{x_0, x_1 \geq 0, x_2 \geq 0, \dots, x_N \geq 0} \{x_0 : e_0 x_0 + \sum_{n=1}^N A_{\bullet n}^* x_n = b^*\}$$

where $c^T \equiv [c_1, \dots, c_N]$, $A \equiv [A_{\bullet 1}, A_{\bullet 2}, \dots, A_{\bullet N}]$ is the original A matrix and

$$(93) A_{\bullet n}^* \equiv \begin{bmatrix} c_n \\ A_{\bullet n} \end{bmatrix} \text{ for } n = 1, \dots, N, b^* \equiv \begin{bmatrix} 0 \\ b \end{bmatrix} \text{ and } e_0 \equiv \begin{bmatrix} 1 \\ 0_M \end{bmatrix}.$$

Define the *super feasibility set* S as the following set in $M+1$ dimensional space:¹¹

$$(94) S \equiv \{z : z = \sum_{n=1}^N A_{\bullet n}^* x_n ; x_1 \geq 0, x_2 \geq 0, \dots, x_N \geq 0\}.$$

The set S is the set of all nonnegative linear combinations of the N column vectors $A_{\bullet n}^*$. Thus S is a set in $M+1$ dimensional space.

A set S is a *cone* if and only if it has the following property:

¹⁰ See Samuelson (1947; 65) and Diewert (1984; 148) for this interpretation for a Lagrange multiplier in classical (equality constrained) optimization theory. Dorfman, Samuelson and Solow (1958; 52-59) have a nice discussion on the meaning and interpretation of dual variables. Problem 16 at the end of the chapter provides an economic interpretation for the dual prices in the general case where conditions (86) hold only as weak inequalities rather than as strict inequalities.

¹¹ Contrast this definition with the feasible set of x 's, which is the set of x such that $x \geq 0_N$ and $Ax = b$.

(95) $z \in S$, $\lambda \geq 0$ implies $\lambda z \in S$.

It is easy to verify that the super feasibility set S defined by (94) is a cone.¹² Note that this set S does not involve the right hand side vector b which appeared in the original constraints for the LP, $Ax = b$. The vector b is often called the *requirements vector*; i.e., a nonnegative vector x must be chosen so that “production” Ax meets the “requirements” b . The *requirements line* L is defined as follows:

$$(96) L \equiv \left\{ \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_M \end{bmatrix} : \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_M \end{bmatrix} = \begin{bmatrix} z_0 \\ b_1 \\ \vdots \\ b_M \end{bmatrix} \text{ where } z_0 \text{ is an unrestricted scalar variable} \right\}.$$

Note that points z that belong to *both* L and S correspond to feasible solutions for the LP (92). An *optimal solution* for (92) will correspond to a point z such that:

- $z \in S$
- $z \in L$ and
- The first component of z is as small as possible subject to $z \in S$ and $z \in L$.

Thus to solve (92), *we look for the lowest point* (with respect to the first component of z) *on the requirements line that also belong to the super feasibility set S* . It also turns out that any hyperplane that supports the set S (is tangent to S) at this lowest point of L provides a set of optimal dual variables. An example will help to illustrate these points.

Suppose that we have 4 production units in a region or a country that uses varying amounts of two inputs and each production unit when run at unit scale produces output that is valued at 1 dollar. The input requirements of the 4 production units when run at unit scale are the $A_{\bullet n}$ listed below:

$$(97) A_{\bullet 1} \equiv \begin{bmatrix} 0 \\ 2 \end{bmatrix}; A_{\bullet 2} \equiv \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}; A_{\bullet 3} \equiv \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}; A_{\bullet 4} \equiv \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Thus the first production unit uses 0 units of the first input and 2 units of the second input in order to produce outputs worth 1 dollar, the second production unit uses 1/2 of a unit of the first input and 1 unit of the second input in order to produce outputs worth 1 dollar and so on. The 4 production units are controlled by a single firm which can allocate $b_1 > 0$ units of input 1 and $b_2 > 0$ units of input 2 across the 4 production units. Production in each plant is subject to constant returns to scale. The firm is interested in allocating the amounts of the two inputs across the 4 plants in order to maximize revenue; i.e., the firm would like to solve the following linear programming problem:

¹² It is also a convex set so it is a convex cone.

$$(98) \max_x \{c^T x : Ax = b ; x \geq 0\}$$

where $c^T = [1,1,1,1]$, $b^T = [b_1, b_2]$, $A = [A_{\bullet 1}, A_{\bullet 2}, A_{\bullet 3}, A_{\bullet 4}]$ and $x^T = [x_1, x_2, x_3, x_4]$ is the vector of nonnegative plant scales.

To put the problem defined by (97) into standard simplex algorithm format, define the $A_{\bullet n}^*$ by adding $-c_n$ as an extra component to the $A_{\bullet n}$:

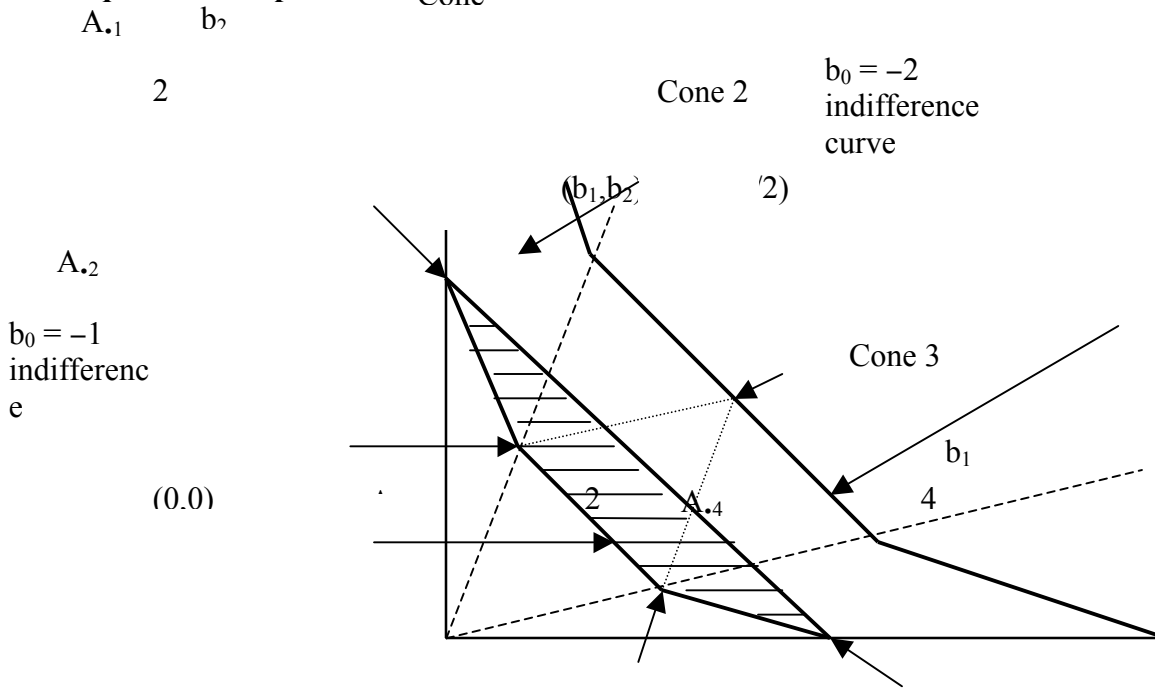
$$(99) A_{\bullet 1}^* = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} ; A_{\bullet 2}^* = \begin{bmatrix} -1 \\ 1/2 \\ 1 \end{bmatrix} ; A_{\bullet 3}^* = \begin{bmatrix} -1 \\ 1 \\ 1/2 \end{bmatrix} ; A_{\bullet 4}^* = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} ; b^* = \begin{bmatrix} 0 \\ b_1 \\ b_2 \end{bmatrix}.$$

Using the above notation, the LP (98) can be rewritten as follows:

$$(100) \max_{x_0, x \geq 0} \{x_0 : e_0 x_0 + A_{\bullet 1}^* x_1 + A_{\bullet 2}^* x_2 + A_{\bullet 3}^* x_3 + A_{\bullet 4}^* x_4 = b^*\}.$$

Now recall the definition of the super feasibility set S , (94). For our example LP (100), this set will be a cone in three dimensional space. Now note that each of the 4 column vectors $A_{\bullet 1}^*$, $A_{\bullet 2}^*$, $A_{\bullet 3}^*$ and $A_{\bullet 4}^*$ will belong to S and the first component for each of these vectors is equal to -1 . Hence if we intersect the cone S with the plane $\{(z_0, z_1, z_2) : z_0 = -1, z_1 \text{ and } z_2 \text{ are unrestricted}\}$, we obtain the shaded set in Figure 2 below.¹³

¹³ A three dimensional diagram would be more instructive but it is more difficult to graph. The reader should think of S as a 3 dimensional set in (b_0, b_1, b_2) space. What we are doing in Figure 2 is intersecting this three dimensional set with various (negative) heights or levels for b_0 and then we plot these level sets in (b_1, b_2) space.



The lower boundary of the shaded set is the counterpart to an indifference curve in consumer theory; this curve gives the set of (b_1, b_2) points which will allow the firm to earn one unit of revenue without wasting resources.¹⁴ The point A_1 is the top point on this curve and it is joined by a straight line to the point A_2 , which in turn is joined to the point A_3 , which in turn is joined to the lowest point on the curve, A_4 . Now suppose that the firm has $(b_1, b_2) = (3/2, 3/2)$ units of the two inputs available to use in the 4 production units. One can go through the algebra of the simplex algorithm in this case and one will find that A_2 and A_3 are the efficient plants to use in this case and the optimal x 's are $x_1^* = 0$, $x_2^* = 1$, $x_3^* = 1$ and $x_4^* = 0$. The optimal objective function in this case is $x_0^* = 2$. In terms of Figure 1, it can be seen that we get to the point $(b_1, b_2) = (3/2, 3/2)$ by moving up the dashed line through A_2 until we hit A_2 and by moving up the dashed line through A_3 until we hit A_3 and then we complete the parallelogram with the dotted lines until we hit $(b_1, b_2) = (3/2, 3/2)$. Note that the two dashed lines enclose a region of (b_1, b_2) that we called *cone 2*. Using the Basis Theorem for linear programs, it can be seen that we will use only production units 2 and 3 provided that (b_1, b_2) falls into this region. Using a similar argument, it can be seen only production units 1 and 2 will be used if (b_1, b_2) falls into the *cone 1* region and only production units 3 and 4 will be used if (b_1, b_2) falls into the *cone 3* region.

Suppose that (b_1, b_2) falls into the interior of the cone 2 region. The corresponding (unique) dual (input) prices can be calculated by solving the following two equations in two unknowns:

$$(101) [1, y_1, y_2]A_2^* = 0; [1, y_1, y_2]A_3^* = 0.$$

¹⁴ In terms of the set S , this curve is the set of (b_1, b_2) points such that $(-1, b_1, b_2)$ belongs to the boundary of S .

Using our data listed in (99), we find that the cone 2 dual solution is:

$$(102) \ y_1^{(2)} = 2/3 ; y_2^{(2)} = 2/3 .$$

Thus the input prices are equal in this case where the resource availability vector falls into the cone 2 region. This corresponds to the fact that the parallel “indifference” curves in this region all have slope equal to -1 (which is equal to $-y_1^{(2)}/y_2^{(2)}$).

Now suppose that (b_1, b_2) falls into the interior of the cone 1 region. The corresponding (unique) dual (input) prices can be calculated by solving the following two equations in two unknowns:

$$(103) \ [1, y_1, y_2]A_{.1}^* = 0 ; [1, y_1, y_2]A_{.2}^* = 0 .$$

Using our data listed in (99), we find that the cone 1 dual solution is:

$$(104) \ y_1^{(1)} = 1 ; y_2^{(1)} = 1/2 .$$

Thus the input 1 price is twice the size of the input 2 price in this region where input 1 is relatively scarce compared to input 2. This corresponds to the fact that the parallel “indifference” curves in this region all have slope equal to -2 (which is equal to $-y_1^{(1)}/y_2^{(1)}$).

Finally suppose that (b_1, b_2) falls into the interior of the cone 3 region. The corresponding (unique) dual (input) prices can be calculated by solving the following two equations in two unknowns:

$$(105) \ [1, y_1, y_2]A_{.3}^* = 0 ; [1, y_1, y_2]A_{.4}^* = 0 .$$

Using our data listed in (99), we find that the cone 3 dual solution is:

$$(106) \ y_1^{(3)} = 1/2 ; y_2^{(3)} = 1 .$$

Thus the input 2 price is twice the size of the input 1 price in this region where input 2 is relatively scarce compared to input 1. This corresponds to the fact that the parallel “indifference” curves in this region all have slope equal to $-1/2$ (which is equal to $-y_1^{(3)}/y_2^{(3)}$).

What happens if the endowment vector (b_1, b_2) happens to fall on the dashed line through the origin and $A_{.2}$ so that (b_1, b_2) is on the boundary of both cone 1 and cone 2? In this case, although the primal solution is unique (x_2^* solves the primal where $A_{.2} x_2^* = b$), the dual solution is not. In this case the dual solution set is:

$$(107) \ (y_1, y_2) = \lambda (y_1^{(1)}, y_2^{(1)}) + (1-\lambda) (y_1^{(2)}, y_2^{(2)}) ; \ 0 \leq \lambda \leq 1 ;$$

i.e., it is the set of all convex combinations of the cone 1 and cone 2 dual prices. Similarly, if the endowment vector (b_1, b_2) happens to fall on the dashed line through the origin and A_3 so that (b_1, b_2) is on the boundary of both cone 2 and cone 3, then in this case, although again the primal solution is unique (x_3^* solves the primal where $A_3 x_3^* = b$), the dual solution is not. In this case the dual solution set is:

$$(108) (y_1, y_2) = \lambda (y_1^{(2)}, y_2^{(2)}) + (1-\lambda) (y_1^{(3)}, y_2^{(3)}) ; 0 \leq \lambda \leq 1 ;$$

i.e., it is the set of all convex combinations of the cone 2 and cone 3 dual prices.

The reader is now encouraged to visualize how the simplex algorithm would work if we had additional “inefficient” production units in the model. If our starting basis matrix had one or more inefficient columns in the basis matrix, then there would exist at least one more efficient column that would lie *below* the dual hyperplane that was generated by the starting columns. Introducing this more efficient column into the basis matrix would lead to a lower dual hyperplane at the next iteration and eventually, the dual hyperplane would lie on or below all of the columns in the model and the algorithm would come to a halt.

The geometry of the simplex algorithm in requirements space is illustrated in Dantzig (1963; 160-165) and Van Slyke (1968).¹⁵ The representation of a linear programming problem in requirements space is a very useful one because we can simultaneously visualize the solution to the primal problem as well as to the dual problem. Also this requirements space geometric approach makes it easy to visualize the comparative statics effects of changing the right hand side vector b .¹⁶

10. The Saddlepoint Criterion for Solving a Linear Program

In this section, we develop some criteria for solving the primal dual problems that do not rely on the simplex algorithm. However, the material in this section is very closely related to the auxiliary result that appeared in the proof of Theorem 3, the duality theorem for linear programs.

Consider the following primal and dual problems:

$$(109) P : \max_x \{c^T x : Ax \leq b ; x \geq 0_N\} ;$$

$$(110) D : \min_y \{y^T b ; y^T A \geq c^T ; y \geq 0_M\}$$

where A is an M by N matrix, b and y are M dimensional vectors and c and x are N dimensional vectors.

Theorem 3 above, the Duality Theorem for Linear Programs, developed optimality criteria for x^* to solve the primal problem (109) and for y^* to solve the dual problem (110), using the mechanics of the simplex algorithm as a tool for establishing the criteria. In

¹⁵ Van Slyke was a student of Dantzig's.

¹⁶ The Figure 1 geometry in activities space is useful for visualizing the comparative statics effects of changing the objective function vector c .

this section, we develop some alternative optimality criteria that do not rely on the simplex algorithm.

In order for x^* to solve P and y^* to solve D, it is obviously necessary that x^* and y^* satisfy the feasibility constraints for their respective problems:

$$(111) \quad Ax^* \leq b ; x^* \geq 0_N ; y^{*T}A \geq c^T ; y^* \geq 0_M .$$

Consider the following condition, which relates the primal and dual objective functions:

$$(112) \quad c^T x^* = y^{*T} b .$$

Theorem 5; Goldman and Tucker (1956): Conditions (111) and (112) are necessary and sufficient for x^* to solve P and for y^* to solve D.

Proof: Recall the auxiliary result that was proven in Theorem 3 above, which may be restated as follows: if x^* is feasible for the primal problem P and y^* is feasible for the dual problem D, then

$$(113) \quad \begin{aligned} c^T x^* &\leq c^T x^* + y^{*T}(b - Ax^*) && \text{since } y^* \geq 0_M \text{ and } Ax^* \leq b \\ &= y^{*T}b + (c^T - y^{*T}A)x^* && \text{rearranging terms} \\ &\leq y^{*T}b && \text{since } x^* \geq 0_N \text{ and } y^{*T}A \geq c^T . \end{aligned}$$

Now suppose that x^* and y^* satisfy (111) and (112). Then repeating the proof of the auxiliary result in Theorem 3 shows that x^* solves P and y^* solves D.

Conversely, let x^* solve P and let y^* solve D. Then x^* and y^* satisfy (111) and (113) and so we have $c^T x^* \leq y^{*T}b$. Suppose $c^T x^* < y^{*T}b$, then we could repeat the proof of Theorem 3 and find a dual solution y^{**} such that $c^T x^* = y^{**T}b < y^{*T}b$. But this last inequality contradicts the assumed optimality of y^* for the dual problem D. Thus our *supposition* is false and $c^T x^* = y^{*T}b$, which is (112). Q.E.D.

Theorem 6; Goldman and Tucker (1956): Consider the following *complementary slackness conditions*:

$$(114) \quad y^{*T}(b - Ax^*) = 0 ;$$

$$(115) \quad (c^T - y^{*T}A)x^* = 0 .$$

The feasibility conditions (111) and the complementary slackness conditions (114) and (115) are necessary and sufficient for x^* to solve P and for y^* to solve D.

Proof: By Theorem 5, we need only show that conditions (114) and (115) are equivalent to condition (112) when conditions (111) hold. But given that (111) holds, we know the inequality (113) holds. But now it can be seen that conditions (114) and (115) are precisely the conditions that are necessary and sufficient to convert the inequality (113) into the equality (112). Q.E.D.

In order to state the next theorem, it is first necessary to define the *Lagrangian* L that corresponds to the primal problem P :

$$(116) L(x,y) \equiv c^T x + y^T [b - Ax].$$

Note that the vector y in (116) plays the role of a vector of *Lagrange multipliers* for the constraints in the primal problem P .

Definition: (x^*, y^*) is a *saddle point* of the Lagrangian L if and only if

$$(117) x^* \geq 0_N; y^* \geq 0_M \text{ and}$$

$$(118) L(x, y^*) \leq L(x^*, y^*) \leq L(x^*, y) \text{ for all } x \geq 0_N \text{ and } y \geq 0_M.$$

Looking at conditions (118), we see that $L(x, y^*)$ attains a *maximum* with respect to x at $x = x^*$ over all $x \geq 0_N$. Conversely, $L(x^*, y)$ attains a *minimum* with respect to y at $y = y^*$ over all $y \geq 0_M$.

Theorem 7; Goldman and Tucker (1956; 77): The saddle point conditions (117) and (118) are necessary and sufficient for x^* to solve P and for y^* to solve D .

Proof: In view of Theorem 6, we need only show that (117) and (118) are equivalent to the feasibility restrictions (111) and the complementary slackness conditions (114) and (115).

Assume that (111), (114) and (115) hold. Conditions (111) imply that conditions (117) hold so we need only show that conditions (118) hold. Using the definition of $L(x, y)$, conditions (118) can be rewritten as follows:

$$(119) c^T x + y^{*T} [b - Ax] \leq c^T x^* + y^{*T} [b - Ax^*] \leq c^T x^* + y^T [b - Ax^*] \\ \text{for all } x \geq 0_N \text{ and } y \geq 0_M.$$

Using (114), the right hand set of inequalities in (119) is equivalent to:

$$(120) 0 \leq y^T [b - Ax^*] \text{ for all } y \geq 0_M.$$

But the feasibility conditions (111) imply that $b - Ax^* \geq 0_M$ so (120) must hold.

Using (115), the left hand set of inequalities in (119) is equivalent to:

$$(121) [c^T - y^{*T} A]x \leq [c^T - y^{*T} A]x^* = 0 \text{ for all } x \geq 0_N.$$

But the feasibility conditions (111) imply that $c^T - y^{*T} A \leq 0_N^T$ so that (121) holds. This completes the proof of the first half of the theorem.

Now assume that the saddle point conditions (117) and (118) hold. The right hand set of inequalities in (118) is equivalent to:

$$(122) \quad y^{*T}[b - Ax^*] \leq y^T[b - Ax^*] \quad \text{for all } y \geq 0_M.$$

But it is easy to see that (122) implies that $b - Ax^* \geq 0_N$. In a similar manner, we can show that the left hand set of inequalities in (118) implies that $c^T - y^{*T}A \leq 0_N^T$. Thus using also (117), we have shown that x^* and y^* satisfy the feasibility conditions (111).

We have shown that $y^* \geq 0_M$ and $b - Ax^* \geq 0_N$. Hence $y^{*T}[b - Ax^*] \geq 0$. Suppose $y^{*T}[b - Ax^*] > 0$. Now set $y = 0_M$ and (122) becomes $0 < y^{*T}[b - Ax^*] \leq 0$, which is a contradiction. Hence our *supposition* is false and $y^{*T}[b - Ax^*] = 0$, which is the complementary slackness condition (114).

Finally, we have shown that $x^* \geq 0_N$ and $c^T - y^{*T}A \leq 0_N^T$. Hence $(c^T - y^{*T}A)x^* \leq 0$. Note that the left hand set of inequalities in (118) or (119) is equivalent to:

$$(123) \quad [c^T - y^{*T}A]x \leq [c^T - y^{*T}A]x^* \quad \text{for all } x \geq 0_N.$$

Now suppose $(c^T - y^{*T}A)x^* < 0$. Set $x = 0_N$ and (123) becomes $0 \leq (c^T - y^{*T}A)x^* < 0$, which is a contradiction. Hence our *supposition* is false and $[c^T - y^{*T}A]x^* = 0$, which is the complementary slackness condition (115). Q.E.D.

The saddle point characterization for optimal solutions to the primal and dual linear programs turns out to be very useful as will be seen in the problems at the end of the chapter.

11. Programming with Variable Coefficients

Consider the following *programming problem with variable coefficients* which has the structure (10) of a standard LP with one exception:

$$(124) \quad \max_{x_0, x_1 \geq 0, x_2 \geq 0, \dots, x_N \geq 0, A_{\cdot 1}^*, \dots, A_{\cdot N}^*} \{x_0 : e_0 x_0 + \sum_{n=1}^N A_{\cdot n}^* x_n = b^* ; A_{\cdot 1}^* \in C^1, \dots, A_{\cdot N}^* \in C^N\}.$$

The only difference between (124) and our standard simplex algorithm primal LP problem (10) is that in (10), the columns $A_{\cdot n}^*$ were fixed but now columns are variable; i.e., $A_{\cdot n}^*$ may be chosen from a closed and bounded set¹⁷ of columns C^n . Obviously, if each column set C^n has only a single member in it, then (124) boils down to our standard LP (10). Dantzig (1963; chapter 22) considered problems of the form (124) where the sets C^n consisted of convex combinations of a finite set of columns. Van Slyke (1968) considered general problems of the type (124).

¹⁷ These restrictions on the sets C^n can be relaxed under certain additional hypotheses. It is sometimes assumed that the sets C^n are convex, but this additional assumption is not necessary for the simplex algorithm to work in this context.

Consider now how to solve a problem of the type (124). Suppose that $A_{.1}^* \in C^1, \dots, A_{.M}^* \in C^M$ exist such that $B = [e_0, A_{.1}^*, \dots, A_{.M}^*]$ is an initial basis matrix for (124); i.e., B^{-1} exists and the last M components of the $M+1$ vector $B^{-1}b^*$ are nonnegative. Consider the following N subproblems:

$$(125) f_n(B) \equiv \min_{A_{.n}^*} \{B_{0.}^{-1} A_{.n}^* : A_{.n}^* \in C^n\}; \quad n = 1, 2, \dots, N.$$

Assuming that we can solve the subproblems (125) (a nontrivial assumption), then the variable coefficients simplex algorithm works as follows:

- (i) If $f_n(B) \geq 0$ for $n = 1, 2, \dots, N$, then the present basis matrix is optimal and the algorithm terminates.
- (ii) If $f_n(B) < 0$ for some n , then there exists $A_{.n}^* \in C^n$ such that $f_n(B) = B_{0.}^{-1} A_{.n}^* < 0$. Introduce $A_{.n}^*$ into the basis matrix using the usual unbounded solutions criterion or the dropping criterion. If we end up in the unbounded solutions criterion case, then the algorithm terminates. Otherwise, we solve the following LP:

$$(125) \max_{x_0, x_1 \geq 0, x_2 \geq 0, \dots, x_M \geq 0, x_n \geq 0} \{x_0 : e_0 x_0 + \sum_{m=1}^M A_{.m}^* x_m + A_{.n}^* x_n = b^*\}$$

and we obtain a new basis matrix. We use this new basis matrix to solve the subproblems (126) and we return to (i) above and repeat the cycle.

We can no longer assert that the simplex algorithm will converge to a solution in a finite number of steps. However, Van Slyke (1968) proved that the algorithm will converge to an optimal solution of (124) under relatively weak regularity conditions.

The above algorithm has been used in the theory of economic planning; see Dorfman, Samuelson and Solow (1958; 59-63) and Malinvaud (1967). The basic idea is the following one. The central planner has a preliminary vector of resource prices $[1, y^{*T}]$, which is transmitted to the various sectors in the economy. Given these prices, the manager of sector n solves a (net) cost minimization problem similar to (126), where $B_{0.}^{-1}$ is replaced by $[1, y^{*T}]$. Each manager then sends back a solution vector $A_{.n}^*$ back to the center. The central planner then solves a simple linear program involving this sectoral data, obtains a new set of prices, transmits this new set of prices to the sectors and so on.¹⁸

The variable coefficients simplex algorithm does depend on the ease with which the subproblems (126) can be solved. If the sets C^n are unit scale production possibilities sets (these sets are dual to a unit cost function or to a unit profit function) and if the dual unit cost or profit functions are available to the programmer, then the subproblems (126) can

¹⁸ Why has this theory of decentralized economic planning not worked in the real world? The assumption of constant returns to scale is somewhat problematic in this theory but probably the main reason it has not worked very well in applications is the *problem of dimensionality*; i.e., real life economic planning problems at the national scale involve thousands if not millions of variables and the resulting programming problems are just too difficult to solve. However, at the level of a firm which has several more or less independent divisions, the above theory of decentralized economic planning could work.

be solved by a simple application of Shephard's (1953) Lemma or Hotelling's (1932) Lemma. In this case, the variable coefficients simplex algorithm is just as easy to use as the ordinary simplex algorithm. Some applications along these lines may be found in Diewert (1975), Diewert and Woodland (1977) and Woodland (1982).

Problems

1. Consider the following linear program:

$$(i) \max_x \{c^T x : A^{(1)}x \leq b^{(1)} ; A^{(2)}x = b^{(2)} ; x \geq 0_N\}.$$

Show that the dual to (i) is:

$$(ii) \min_{y^{(1)}, y^{(2)}} \{y^{(1)T}b^{(1)} + y^{(2)T}b^{(2)} : y^{(1)T}A^{(1)} + y^{(2)T}A^{(2)} \geq c^T ; y^{(1)} \geq 0_{M_1} ; y^{(2)} \text{ unrestricted}\}.$$

Hint: Write $A^{(2)}x = b^{(2)}$ as $A^{(2)}x \leq b^{(2)}$ and $-A^{(2)}x \leq -b^{(2)}$. *Comment:* If $M_1 = 0$, then the primal problem has equality constraints and the corresponding dual variables are unrestricted.

2. Consider the following linear program:

$$(i) \max_{x^{(1)}, x^{(2)}} \{c^{(1)T}x^{(1)} + c^{(2)T}x^{(2)} : A^{(1)}x^{(1)} + A^{(2)}x^{(2)} \leq b ; x^{(1)} \geq 0_{N_1} ; x^{(2)} \text{ unrestricted}\}.$$

Show that the dual to (a) is

$$(ii) \min_y \{y^T b : y^T A^{(1)} \geq c^{(1)T} ; y^T A^{(2)} = c^{(2)T} ; y \geq 0_M\}.$$

3. Consider a firm that has 4 production processes (or activities) that can be used to produce two outputs.¹⁹ Each process uses one primary input (call it labour) and produces one of the two outputs. Some processes use the other output as an intermediate input. The firm wishes to minimize the aggregate labour cost of producing given positive amounts, $b_1 > 0$ and $b_2 > 0$, of the two outputs. The linear program that the firm wishes to solve is:

$$(i) \min_{x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0} \{2x_1 + 1x_2 + 1x_3 + 2x_4 : \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \geq \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\}.$$

¹⁹ Each process can be thought of as a Leontief production function that does not allow for substitution between inputs. Leontief (1936) (1941) was a pioneer in setting up production models of entire economies using Leontief type production functions. His influence lives on today: many national statistical agencies produce input-output tables based on his model of production.

The first two activities produce the finally demanded commodity b_1 while the last two activities produce the finally demanded commodity b_2 .

(a) Solve the above LP when $b_1 = 1$ and $b_2 = 1$. *Hint:* rewrite (i) in standard simplex algorithm format as follows:

$$(ii) \max_{x_0, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, s_1 \geq 0, s_2 \geq 0} \left\{ x_0 : \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_0 + \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} x_2 + \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} x_3 + \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} x_4 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} s_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} s_2 = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} \right\}.$$

Note that the x_1 and x_4 columns can be used in a starting basis matrix.

The dual problem to (i) is:

$$(iii) \max_{y_1 \geq 0, y_2 \geq 0} \{ y_1 b_1 + y_2 b_2 : [y_1, y_2]^T \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 1 \end{bmatrix} \leq [2, 1, 1, 2] \}.$$

(b) Find the dual prices y_1^* and y_2^* for problem (iii) when $b_1 = 1$ and $b_2 = 1$. *Hint:* You need only list the last two components of B_0^{-1} where B is the optimal basis matrix for (ii).

(c) Finally, solve the LP (i) for a general $b_1 > 0$ and $b_2 > 0$ and show that the corresponding dual prices do not depend on b_1 and b_2 . *Comment:* This problem provides a concrete illustration of the Samuelson (1951) Nonsubstitution Theorem; i.e., the optimal activities do not depend on the particular vector of final demands $[b_1, b_2]$ that we are required to produce in this single input (labour theory of value) economy and neither do the dual prices, which can be interpreted as equilibrium prices for the outputs.

4. Consider the following *zero sum game theory problem*, where we list the payoff matrix of the student:

		<i>Student's Strategy Space</i>	
		Go to lecture	Sleep in
<i>Professor's Strategy Space</i>	Give a good lecture	2	1
	Give a terrible lecture	0	3

We suppose that the student wishes to choose a strategy, which consists of the probability of going to a lecture, $p_1 \geq 0$, and the probability of sleeping in, $p_2 \geq 0$, where $p_1 + p_2 = 1$. We further suppose that the somewhat psychotic student suspects that the professor is perversely attempting to minimize the student's payoff. Thus the student would be wise to choose a strategy which would ensure him or her a minimum amount of utility

irrespective of the professor's actions; this is called a *minimax strategy*. In order to find the minimax strategy, the student will want to solve the following linear program:

$$(i) \max_{v \geq 0, p_1 \geq 0, p_2 \geq 0} \{v : v \leq 2p_1 + 1p_2 ; v \leq 0p_1 + 3p_2 ; 1 = p_1 + p_2\}.$$

Since the student's payoff matrix is nonnegative, we can define new variables x_1 and x_2 which will also be nonnegative as follows:

$$(ii) x_1 \equiv p_1/v ; x_2 \equiv p_2/v.$$

Note that $p_1 + p_2 = vx_1 + vx_2 = 1$ so v can be expressed in terms of x_1 and x_2 as follows:

$$(iii) v = 1/(x_1 + x_2).$$

Thus maximizing v is equivalent to minimizing $x_1 + x_2$ and we find that the LP (i) is equivalent to the following LP:

$$(iv) \min_{x_1 \geq 0, x_2 \geq 0} \{x_1 + x_2 : 2x_1 + 1x_2 \geq 1 ; 0x_1 + 3x_2 \geq 1\}.$$

(a) Solve the LP (iv) and use (ii) and (iii) to calculate the student's optimal strategy, p_1^* and p_2^* .

(b) What probability mix of good and terrible lectures should the professor use to ensure a minimum payoff to the student? *Hint*: We can transform the professor's minimization problem into the dual of the LP (iv). *Comment*: This problem provides an example of the *minimax theorem* for zero sum two person games. *Reference*: Dorfman, Samuelson and Solow (1958; chapters 15-16).

5. *Complementary Slackness in Linear Programming* ; Dantzig (1963; 136): Consider the following primal LP:

$$(i) \max_{x \geq 0_N} \{c^T x : Ax \leq b\} \text{ where } A \text{ is an } M \text{ by } N \text{ matrix.}$$

Suppose a finite optimal solution $x^0 \geq 0_N$ exists for the above problem. Then by the *duality theorem for linear programs*, an optimal vector of dual variables $y^{0T} \equiv [y_1^0, \dots, y_M^0] \geq 0_M^T$ will also exist.

(a) Show that if the m th constraint in the primal LP is satisfied with a strict inequality when evaluated at the primal optimal solution x^0 so that $A_{m \cdot} x^0 < b_m$, then the corresponding dual variable y_m^0 must equal 0. *Hint*: If the m th constraint is satisfied with a strict inequality, then the corresponding slack variable must be in the optimal basis matrix.

(b) Give an economic interpretation of the above result.

(c) Prove that $y^{0T}[b - Ax^0] = 0$ and $[c^T - y^{0T}A]x^0 = 0$.

6. *Farkas (1902) Lemma*: Let A be a nonzero M by N matrix and c be an N dimensional vector. Then, *either* there exists an x such that

(i) $Ax \leq 0_M$ and $c^T x > 0$

or there exists a y such that

(ii) $y^T A = c^T$ and $y \geq 0_M$.²⁰

Prove the above result. *Hint*: Consider the following primal and dual problems:

(iii) $\max_x \{c^T x : Ax \leq 0_M\}$;

(iv) $\min_y \{y^T 0_M : y^T A = c^T ; y \geq 0_M\}$.

Using the results of Problems 1 and 2 above, it can be seen that (iv) is the dual to (iii). Since $x = 0_N$ is feasible for (iii), there are only two cases that can occur for (iii): *Case 1*: the optimal objective function for (iii) becomes unbounded and *Case 2*: (iii) has a finite optimal solution. Show that Case 1 corresponds to (i) and Case 2 corresponds to (ii). *Comment*: Farkas' Lemma can be used to prove Theorem 3, the duality theorem for linear programs (this is the path followed by Tucker (1956)) and as the above problem shows, the duality theorem established via the simplex algorithm can be used to prove Farkas' Lemma (this is the path followed by Dantzig (1963; 145)). Mangasarian (1969; chapter 2) calls Farkas' Lemma a *theorem of the alternative* and he lists several other similar theorems, which we will give as problems 7-9 below. These theorems of the alternative turn out to be very useful in the theory of nonlinear programming and in the theory of economic policy. In particular, theorems of the alternative can be used to establish the existence of Pareto improving changes in economic policy instruments. For examples of these applications to the theory of economic policy, see Diewert (1978; 281) (1983) (1987), Weymark (1979) and Diewert, Turunen-Red and Woodland (1989) (1991).

7. *Motzkin's (1936) Transposition Theorem*; see Mangasarian (1969; 28-29): Let E be an M_1 by N matrix, F be an M_2 by N matrix and G be an M_3 by N matrix where $M_1 > 0$, $N > 0$, $M_2 \geq 0$ and $M_3 \geq 0$.²¹ Then *either* there exists x such that

(i) $Ex >> 0_{M_1}$; $Fx \geq 0_{M_2}$; $Gx = 0_{M_3}$

or there exist y^1, y^2, y^3 such that

(ii) $y^{1T}E + y^{2T}F + y^{3T}G = 0_N^T$; $y^1 > 0_{M_1}$; $y^2 \geq 0_{M_2}$; y^3 unrestricted.

Hint: Consider the following LP which is closely related to (i):

²⁰ Both (i) and (ii) cannot hold.

²¹ If $M_2 = 0$, then drop the matrix F from the problem; if $M_3 = 0$, then drop the matrix G from the problem.

$$(iii) \max_x \{0_N^T x : -Ex \leq -1_{M_1}; -Fx \leq 0_{M_2}; -Gx = 0_{M_3}; x \text{ unrestricted}\}.$$

The dual to (iii) turns out to be the following LP, which is closely related to (ii):

$$(iv) \min_{y^1, y^2, y^3} \{-y^{1T} 1_{M_1} : y^{1T} E + y^{2T} F + y^{3T} G = 0_N^T; y^1 \geq 0_{M_1}; y^2 \geq 0_{M_2}; y^3 \text{ unrestricted}\}.$$

Note that the objective function for (iii) is bounded. Hence, there are two cases that can occur for (iii): Case 1: a feasible solution for (iii) exists and hence we have a bounded optimal solution for (iii) or Case 2: no feasible solution for (iii) exists.

8. *Tucker's (1956; 14) Theorem of the Alternative*; see Mangasarian (1969; 29): Let B be an M_1 by N matrix, C be an M_2 by N matrix and D be an M_3 by N matrix where $M_1 > 0$, $N > 0$, $M_2 \geq 0$ and $M_3 \geq 0$. Then *either* there exists x such that

$$(i) Bx > 0_{M_1}; Cx \geq 0_{M_2}; Dx = 0_{M_3}$$

or there exist y^1, y^2, y^3 such that

$$(ii) y^{1T} B + y^{2T} C + y^{3T} D = 0_N^T; y^1 \gg 0_{M_1}; y^2 \geq 0_{M_2}; y^3 \text{ unrestricted}.$$

Hint: Consider the following LP which is closely related to (i):

$$(iii) \max_x \{1_{M_1}^T B x : -Bx \leq 0_{M_1}; -Cx \leq 0_{M_2}; -Dx = 0_{M_3}; x \text{ unrestricted}\}.$$

The dual to (iii) turns out to be the following LP, which is closely related to (ii):

$$(iv) \min_{y^1, y^2, y^3} \{y^{1T} 0_{M_1} + y^{2T} 0_{M_2} + y^{3T} 0_{M_3} : y^{1T} B + y^{2T} C + y^{3T} D = -1_{M_1}^T B; \\ y^1 \geq 0_{M_1}; y^2 \geq 0_{M_2}; y^3 \text{ unrestricted}\}.$$

Note that $x = 0_N$ is always feasible for (iii). Hence, there are two cases that can occur for (iii): Case 1: a feasible solution for (iii) exists but the corresponding optimal objective function is 0 and hence we have a bounded optimal solution for (iii) or Case 2: a feasible solution for (iii) exists and the objective function when evaluated at this feasible solution is positive and so in this case we obtain an unbounded solution for the primal and no feasible solution for the dual (iv).

9. *Slater's (1951) Theorem of the Alternative*: Let A be M_1 by N, B be M_2 by N, C be M_3 by N and D be M_4 by N with $M_1 > 0$, $M_2 > 0$ and $N > 0$. Then *either*

$$(i) Ax \gg 0_{M_1}; Bx > 0_{M_2}; Cx \geq 0_{M_3}; Dx = 0_{M_4} \text{ has a solution } x \text{ or}$$

$$(ii) y^{1T} A + y^{2T} B + y^{3T} C + y^{4T} D = 0_N^T \\ \text{with } y^1 > 0_{M_1}; y^2 \geq 0_{M_2}; y^3 \geq 0_{M_3} \text{ or}$$

$$y^1 \geq 0_{M_1}; y^2 \gg 0_{M_2}; y^3 \geq 0_{M_3}$$

has a solution y^1, y^2, y^3, y^4 .²²

Hint: Consider the following LP which is closely related to (i):

$$(iii) \max_x \{ 1_{M_1}^T B x : -Ax \leq -1_{M_1}; -Bx \leq 0_{M_2}; -Cx \leq 0_{M_3}; -Dx = 0_{M_4}; x \text{ unrestricted} \}.$$

The dual to (iii) turns out to be the following LP, which is closely related to (ii):

$$(iv) \min_{y^1, y^2, y^3, y^4} \{ y^{1T} 1_{M_1} + y^{2T} 0_{M_2} + y^{3T} 0_{M_3} + y^{4T} 0_{M_4} : y^{2T} B + y^{2T} B + y^{3T} C + y^{4T} D \\ = -1_{M_1}^T B; y^1 \geq 0_{M_1}; y^2 \geq 0_{M_2}; y^2 \geq 0_{M_3}; y^4 \text{ unrestricted} \}.$$

For this problem, there are three cases to analyze: Case 1: a finite optimal solution for (iii) exists; Case 2: (iii) has an unbounded solution and Case 3: (iii) has no feasible solution.

10. *Factor Price Equalization Theorem*; Samuelson (1958), Diewert and Woodland (1977): Suppose that a country has N production functions of the Leontief no substitution variety where each of the N activities uses varying amounts of M primary inputs. Suppose that each production function produces a single output and suppose that all outputs are traded at the positive world price vector $p = [p_1, p_2, \dots, p_N]^T$. Suppose further that the economy has an endowment of M primary inputs, $v = [v_1, v_2, \dots, v_M]^T \gg 0_M$. The M by N matrix of unit output input requirements is $A = [a_{mn}]$ where each $a_{mn} \geq 0$ for $m = 1, \dots, M$ and $n = 1, \dots, N$. We assume that producers take output prices as fixed and they collectively attempt to maximize the value of their country's output subject to their country's M primary resource constraints; i.e., they attempt to solve the following LP (there are no intermediate inputs in this simple model):

$$(i) \max_x \{ p^T x : Ax \leq v; x \geq 0_N \}$$

where x is a nonnegative vector of industry outputs. The dual to (i) is:

$$(ii) \min_w \{ w^T v : w^T A \geq p^T; w \geq 0_M \}$$

where w can be interpreted as vector of primary input prices or resource costs. Thus in the dual problem, we attempt to choose input prices that will minimize the aggregate cost of production subject to the constraints that nonpositive profits are made in each industry. By the complementary slackness conditions for linear programs, we know that for industries that are operated at a positive scale in a solution to (i), then 0 profits will be made in that industry in equilibrium. Industries that are operated at 0 scale will make negative or 0 profits at the equilibrium input prices. Thus if producers collectively solve the above programming problems, then they will have chosen output levels that

²² Both (i) and (ii) cannot hold.

maximize the country's value of output at international prices and they will have also determined competitive input prices for their country.

Consider the following example, which has 3 production activities ($N = 3$) and two primary inputs ($M = 2$):

$$(iii) \quad \max_{x_0, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, s_1 \geq 0, s_2 \geq 0}$$

$$\begin{aligned} & \max_{x_0, x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, s_1 \geq 0, s_2 \geq 0} \left\{ x_0 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} x_0 + \begin{bmatrix} -p_1 \\ 2 \\ 0 \end{bmatrix} x_1 + \begin{bmatrix} -p_2 \\ 0 \\ 2 \end{bmatrix} x_2 \right. \\ & \left. + \begin{bmatrix} -p_3 \\ 3/2 \\ 1 \end{bmatrix} x_3 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} s_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} s_2 = \begin{bmatrix} 0 \\ v_1 \\ v_2 \end{bmatrix} \right\}. \end{aligned}$$

(a) Suppose $p_1 = p_2 = p_3 = 1$. Calculate the optimal dual prices to the LP (iii), w_1^* and w_2^* , and show that they do not depend on the particular values that $v_1 > 0$ and $v_2 > 0$ take on. *Hint:* Apply the Basis Theorem for Linear Programs. *Comment:* If each country in the world had access to the same technology matrix and faced the same vector of world prices²³, then each country which had an endowment vector which was a nonnegative linear combination of the same activity vectors A_n^* in an optimal basis matrix would have the same vector of factor prices w^* . Thus there is a tendency for trade in commodities to equalize factor prices across countries. The above result can be generalized to deal with classical production functions and the existence of intermediate inputs; see Samuelson (1958), Diewert and Woodland (1977) and Woodland (1982).

11. Anderson (1958; 349): Let $A(\alpha) \equiv [a_{mn}(\alpha)]$ be an M by M matrix whose elements depend on a scalar parameter α . Suppose when $\alpha = \alpha^0$, $[A(\alpha^0)]^{-1} \equiv B(\alpha^0)$ exists. Let the matrix of derivatives of $A(\alpha)$ evaluated at α^0 be denoted as $A'(\alpha^0) \equiv [da_{mn}(\alpha^0)/d\alpha]$ and let the matrix of derivatives of $B(\alpha)$ evaluated at α^0 be denoted as $B'(\alpha^0) \equiv [db_{mn}(\alpha^0)/d\alpha]$. Show that

$$(i) \quad B'(\alpha^0) = -[A(\alpha^0)]^{-1} A'(\alpha^0) [A(\alpha^0)]^{-1}.$$

Hint: Differentiate both sides of the matrix equation $A(\alpha^0)[A(\alpha^0)]^{-1} = I_M$ with respect to α .

12. *The Comparative Statics of the Strict Complementary Slackness Case:* Consider the following primal LP where the elements of the c and b vectors depend continuously on a scalar parameter α as do the elements of the A matrix, which is M by N where $M < N$:

²³ Hence we are ignoring transport costs. We are also assuming that each producer behaves competitively so there are a lot of assumptions in this model that are unlikely to be completely satisfied in the real world. Moreover, factor price equalization becomes increasingly unlikely as the number of primary inputs (and domestically produced services which are not internationally traded) increases. But the factor price equalization theorem is a "pretty" theoretical result!

$$(i) V(\alpha) \equiv \max_x \{c(\alpha)^T x : A(\alpha)x = b(\alpha) ; x \geq 0_N\}.$$

Define the square M by M matrix, $\tilde{A}(\alpha^0) \equiv [A_{\cdot 1}(\alpha^0), A_{\cdot 2}(\alpha^0), \dots, A_{\cdot M}(\alpha^0)]$ and we suppose that this matrix satisfies the following conditions:

$$(ii) [\tilde{A}(\alpha^0)]^{-1} \text{ exists ;}$$

$$(iii) \tilde{x}(\alpha^0) \equiv [\tilde{A}(\alpha^0)]^{-1}b(\alpha^0) >> 0_M ; x(\alpha^0)^T \equiv [\tilde{x}(\alpha^0)^T, 0_{N-M}^T] ;$$

$$(iv) y(\alpha^0)^T \equiv [c_1(\alpha^0), c_2(\alpha^0), \dots, c_M(\alpha^0)][\tilde{A}(\alpha^0)]^{-1} \equiv \tilde{c}(\alpha^0)^T [\tilde{A}(\alpha^0)]^{-1} ;$$

$$(v) y(\alpha^0)^T A_{\cdot n}(\alpha^0) > c_n(\alpha^0) \text{ for } n = M+1, M+2, \dots, N.$$

From our knowledge of the simplex algorithm, it can be seen that conditions (ii)-(v) are sufficient to imply that the first M columns of the $A(\alpha^0)$ matrix form an optimal basis matrix for the LP defined by (i) when $\alpha = \alpha^0$. Conditions (v) imply that the $x(\alpha^0)$ solution defined in (iii) is the *unique solution* to the primal when $\alpha = \alpha^0$ and the strict inequalities in (iii) along with assumption (ii) implies that $y(\alpha^0)$ defined in (iv) is the *unique solution* to the following dual problem when $\alpha = \alpha^0$:

$$(vi) V(\alpha) \equiv \min_y \{y^T b(\alpha) : y^T A(\alpha) \geq c(\alpha) ; y \text{ unrestricted}\}.$$

The continuity of the function $A(\alpha)$, $b(\alpha)$ and $c(\alpha)$ means that for α sufficiently close to α^0 , the primal solution $x(\alpha)$ to $V(\alpha)$ defined by (i) will be

$$(vii) \tilde{x}(\alpha) \equiv [\tilde{A}(\alpha)]^{-1}b(\alpha) ; x(\alpha)^T \equiv [\tilde{x}(\alpha)^T, 0_{N-M}^T]$$

and the dual solution $y(\alpha)$ to $V(\alpha)$ defined by (vi) will be

$$(viii) y(\alpha)^T \equiv [c_1(\alpha), c_2(\alpha), \dots, c_M(\alpha)][\tilde{A}(\alpha)]^{-1} \equiv \tilde{c}(\alpha)^T [\tilde{A}(\alpha)]^{-1}.$$

(a) Show that if $A(\alpha)$ and $b(\alpha)$ are differentiable at $\alpha = \alpha^0$, then the optimal x solution to (i) is differentiable at $\alpha = \alpha^0$, with

$$(ix) x'(\alpha^0)^T \equiv [\tilde{x}'(\alpha^0)^T, 0_{N-M}^T] \text{ and}$$

$$(x) \tilde{x}'(\alpha^0) \equiv [\tilde{A}(\alpha^0)]^{-1}b'(\alpha^0) - [\tilde{A}(\alpha^0)]^{-1} \tilde{A}'(\alpha^0) [\tilde{A}(\alpha^0)]^{-1}b(\alpha^0).$$

(b) Show that if $A(\alpha)$ and $c(\alpha)$ are differentiable at $\alpha = \alpha^0$, then the optimal y solution to (vi) is differentiable at $\alpha = \alpha^0$, with

$$(xi) y'(\alpha)^T \equiv \tilde{c}'(\alpha)^T [\tilde{A}(\alpha)]^{-1} - \tilde{c}(\alpha)^T [\tilde{A}(\alpha^0)]^{-1} \tilde{A}'(\alpha^0) [\tilde{A}(\alpha^0)]^{-1}.$$

(c) Show that if $A(\alpha)$, $b(\alpha)$ and $c(\alpha)$ are differentiable at $\alpha = \alpha^0$, then the optimal primal and dual objective functions $V(\alpha)$ defined by (i) or (vi) are differentiable at $\alpha = \alpha^0$ with

$$(xii) V'(\alpha) = \tilde{c}'(\alpha)^T [\tilde{A}(\alpha)]^{-1} b(\alpha^0) - \tilde{c}(\alpha)^T [\tilde{A}(\alpha^0)]^{-1} \tilde{A}'(\alpha^0) [\tilde{A}(\alpha^0)]^{-1} b(\alpha^0) \\ + \tilde{c}(\alpha)^T [\tilde{A}(\alpha)]^{-1} b'(\alpha^0).$$

(d) Specialize the results in (a)-(c) to the case where only A depends on α ; i.e., assume b and c are constant vectors.

(e) Specialize the results in (a)-(c) to the case where only b depends on α .

(f) Specialize the results in (a)-(c) to the case where only c depends on α .

Comment: For similar comparative statics results for a general nonlinear programming problem, see Diewert (1984).

13. *Some Global Comparative Statics Results for Linear Programming Problems;* Beckmann (1955-56): Consider the following two linear programs:

$$(i) \max_x \{c^j x : A^j x \leq b^j ; x \geq 0_N\} ; \quad j = 0, 1$$

where A^j is an M by N matrix. The dual problems that correspond to the primal problems (i) are:

$$(ii) \min_y \{y^T b^j : y^T A^j \geq c^j ; y \geq 0_M\} ; \quad j = 0, 1.$$

Suppose that the problems in (i) and (ii) have finite optimal solutions, x^j and y^j respectively when $j = 0, 1$. The changes ΔA , Δb , Δc , Δx and Δy are defined as follows:

$$(iii) A^1 \equiv A^0 + \Delta A ; b^1 \equiv b^0 + \Delta b ; c^1 \equiv c^0 + \Delta c ; x^1 \equiv x^0 + \Delta x ; y^1 \equiv y^0 + \Delta y .$$

(a) Show that the changes ΔA , Δb , Δc , Δx and Δy and x^0 and y^0 satisfy the following inequality:

$$(iv) [\Delta c^T - y^{0T} \Delta A] \Delta x - \Delta y^T [\Delta b - \Delta A x^0] \geq 0.$$

Hint: Use the Saddlepoint inequalities (118) for the LP's in (i).

(b) Specialize (iv) to the case where only b changes. Further specialize (iv) to the case where only the first component of b changes. Provide an economic interpretation for the resulting formula.

(c) Specialize (iv) to the case where only c changes. Further specialize (iv) to the case where only the first component of c changes. Provide an economic interpretation for the resulting formula.

(d) Specialize (iv) to the case where only one component of A changes, say a_{mn} changes.

(e) *Bailey's (1955-56) Inequalities*: Show that under the above conditions, the following two inequalities are also valid:²⁴

$$\begin{aligned}
 \text{(v)} \quad \Delta[c^T - y^T A] \Delta x &\equiv [(c^{1T} - y^{1T} A^1) - (c^{0T} - y^{0T} A^0)] \Delta x \\
 &= [\Delta c^T - y^{1T} \Delta A^1 + y^{0T} \Delta A^0] \Delta x \\
 &= [\Delta c^T - \Delta y^T A^1 - y^{0T} \Delta A] \Delta x \\
 &= [\Delta c^T - y^{1T} \Delta A - \Delta y^T A^0] \Delta x \\
 &\geq 0.
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad \Delta y^T \Delta[b - Ax] &\equiv \Delta y^T [(b^1 - A^1 x^1) - (b^0 - A^0 x^0)] \\
 &= \Delta y^T [\Delta b - A^1 \Delta x^1 + A^0 \Delta x^0] \\
 &= \Delta y^T [\Delta b - \Delta A x^1 - A^0 \Delta x] \\
 &= \Delta y^T [\Delta b - A^1 \Delta x^1 - \Delta A x^0] \\
 &\leq 0.
 \end{aligned}$$

Hint: Define the slack vectors for the primal problems as $s^j \equiv b^j - A^j x^j \geq 0_M$ for $j = 0, 1$. Recall that $y^{iT} s^j = 0$ for $j = 0, 1$. Now note that $\Delta y^T \Delta[b - Ax] = [y^{1T} - y^{0T}] [s^1 - s^0] = y^{1T} s^1 + y^{0T} s^0 - y^{1T} s^0 - y^{0T} s^1$. Using the complementary slackness conditions for each primal LP, the inequality (vi) follows readily. A similar analysis works for (v).

14. Define the primal and dual LP's for two c vectors as follows:

$$\begin{aligned}
 \text{(i)} \quad V(c^j) &\equiv \max_x \{c^{jT} x : Ax \leq b ; x \geq 0_N\} ; & j = 1, 2 ; \\
 \text{(ii)} \quad V(c^j) &\equiv \min_y \{y^T b : y^T A \geq c^{jT} ; y \geq 0_M\} ; & j = 1, 2 .
 \end{aligned}$$

Suppose that feasible solutions exist for both the primal and dual problems defined by (i) and (ii) so that each problem has a finite optimal solution. Let λ be a scalar such that $0 < \lambda < 1$ and define the following LP:

$$\text{(iii)} \quad V(\lambda c^1 + (1-\lambda)c^2) \equiv \max_x \{[\lambda c^1 + (1-\lambda)c^2]^T x : Ax \leq b ; x \geq 0_N\} .$$

(a) Show that a finite optimal solution for the LP defined by (iii) exists. *Hint*: Show that a feasible solution exists for (iii) and its dual.

(b) Show that:

$$\text{(iv)} \quad V(\lambda c^1 + (1-\lambda)c^2) \leq \lambda V(c^1) + (1-\lambda)V(c^2) .$$

Comment: This problem shows that the optimized objective function of a linear programming problem is a *convex function* in the c vector.

15. Define the primal and dual LP's for two b vectors as follows:

²⁴ These two inequalities imply Beckmann's inequalities.

$$\begin{aligned} \text{(i)} \quad V(b^j) &\equiv \max_x \{c^T x : Ax \leq b^j ; x \geq 0_N\} ; & j = 1, 2 ; \\ \text{(ii)} \quad V(b^j) &\equiv \min_y \{y^T b^j : y^T A \geq c^T ; y \geq 0_M\} ; & j = 1, 2 . \end{aligned}$$

Suppose that feasible solutions exist for both the primal and dual problems defined by (i) and (ii) so that each problem has a finite optimal solution. Let λ be a scalar such that $0 < \lambda < 1$ and define the following LP:

$$\text{(iii)} \quad V(\lambda b^1 + (1-\lambda)b^2) \equiv \max_x \{c^T x : Ax \leq \lambda b^1 + (1-\lambda)b^2 ; x \geq 0_N\} .$$

(a) Show that a finite optimal solution for the LP defined by (iii) exists. *Hint:* Show that a feasible solution exists for (iii) and its dual.

(b) Show that:

$$\text{(iv)} \quad V(\lambda b^1 + (1-\lambda)b^2) \geq \lambda V(b^1) + (1-\lambda)V(b^2) .$$

Comment: This problem shows that the optimized objective function of a linear programming problem is a *concave function* in the right hand side vector b over the set of b 's for which there is a feasible solution for the primal problem.

16. Consider the LP defined by (99) and (100) and suppose that $[b_1^*, b_2^*] = [2, 1]$ so that the b vector is a multiple of $A_{.3}$ and b lies on the dashed line through $A_{.3}$ in Figure 2. Now suppose b_1 *increases* by a marginal unit from its initial value so that the new $[b_1, b_2^*]$ lies in the interior of the cone 3 region. Regard the optimized objective function of the LP defined by (100) as a function of b_1 , say $V(b_1)$.

(a) Show that for $[b_1, b_2^*]$ in the interior of the cone 3 region, the derivative of the optimal objective function with respect to b_1 is

$$\text{(i)} \quad dV(b_1)/db_1 = y_1^{(3)} = 1/2 .$$

Hint: Recall (106) and use part (xii) of Problem 12 above.

Now suppose b_1 *decreases* by a marginal unit from its initial value of $[b_1^*, b_2^*] = [2, 1]$ so that the new $[b_1, b_2^*]$ lies in the interior of the cone 2 region. Again, regard the optimized objective function of the LP defined by (100) as a function of b_1 , say $V(b_1)$.

(b) Show that for $[b_1, b_2^*]$ in the interior of the cone 2 region, the derivative of the optimal objective function with respect to b_1 is

$$\text{(ii)} \quad dV(b_1)/db_1 = y_1^{(2)} = 2/3 .$$

Hint: Recall (102) and use part (xii) of Problem 12 above.

(c) Use parts (a) and (b) above to show that

- (iii) $dV^+(b_1^*)/db_1 \equiv \lim_{h \rightarrow 0, h > 0} [V(b_1^* + h) - V(b_1^*)]/h = y_1^{(3)}$ and
 (iv) $dV^-(b_1^*)/db_1 \equiv \lim_{h \rightarrow 0, h < 0} [V(b_1^* + h) - V(b_1^*)]/h = y_1^{(2)}$.

Comment: This problem shows that for the LP defined by (100), the optimized objective function is not always differentiable with respect to the components of the right hand side b vector. However, the *one sided derivatives* or *directional derivatives* of $V(b)$ always exist. This problem also casts a bit more light on the meaning of the dual prices in the general case (as opposed to the special situation that we had in problem 12 where the dual price vector was locally unique).

References

- Anderson, T.W. (1958), *An Introduction to Multivariate Statistical Analysis*, New York: John Wiley.
- Baily, M.J. (1955-6), "A Generalized Comparative Statics in Linear Programming", *Review of Economic Studies* 23, 236-240.
- Beale, E.M.L. (1955), "Cycling in the Dual Simplex Algorithm", *Naval Research Logistics Quarterly* 2, 269-276.
- Beckmann, M.J. (1955-6), "Comparative Statics in Linear Programming and the Giffin Paradox", *Review of Economic Studies* 23, 232-235.
- Carathéodory, C. (1911), "Über den Variabilitätsbereich der Fourierschen Konstanten von positiven harmonischen Funktionen", *Rendiconti Circolo Matematico Palermo* 32, 193-217.
- Dantzig, G.B. (1951), "Maximization of a Linear Function of Variables subject to Linear Inequalities", pp.339-347 in *Activity Analysis of Production and Allocation*, T.C. Koopmans (ed.), New York: John Wiley.
- Dantzig, G.B. (1963), *Linear Programming and Extensions*, Princeton N.J.: Princeton University Press.
- Diewert, W.E. (1975), "The Samuelson Nonsubstitution Theorem and the Computation of Equilibrium Prices", *Econometrica* 43, 57-64.
- Diewert, W.E. (1978), "Optimal Tax Perturbations", *Journal of Public Economics* 10, 138-177.
- Diewert, W.E. (1983), "Cost-Benefit Analysis and Project Evaluation: A Comparison of Alternative Approaches", *Journal of Public Economics* 22, 265-302.
- Diewert, W.E. (1984), "Sensitivity Analysis in Economics", *Computers and Operations Research* 11:2, 141-156.

- Diewert, W.E. (1987), "The Effects of an Innovation: A Trade Theory Approach", *The Canadian Journal of Economics* 20, 694-714.
- Diewert, W.E., A.H. Turunen-Red and A.D. Woodland (1989), "Productivity and Pareto Improving Changes in Taxes and Tariffs", *Review of Economic Studies* 56, 199-216.
- Diewert, W.E., A.H. Turunen-Red and A.D. Woodland (1991), "Tariff Reform in a Small Open Multi-Household Economy with Domestic Distortions and Nontraded Goods", *International Economic Review* 32, 937-957.
- Diewert, W.E. and A.D. Woodland (1977), "Frank Knight's Theorem in Linear Programming Revisited", *Econometrica* 45, 375-398.
- Dorfman, R., P.A. Samuelson and R.M. Solow (1958), *Linear Programming and Economic Analysis*, New York: McGraw-Hill Book Company.
- Farkas, J. (1902), "Über die Theorie der einfachen Ungleichen", *Journal für reine und angewandte Mathematik* 124, 1-24.
- Fenchel, W. (1953), "Convex Cones, Sets and Functions", Lecture Notes at Princeton University, Department of Mathematics, Princeton, N.J.
- Goldman, A.J. and A.W. Tucker (1956), "Theory of Linear Programming", pp. 53-97 in *Linear Inequalities and Related Systems*, H.W. Kuhn and A.W. Tucker (eds.), Princeton N.J.: Princeton University Press.
- Hoffman, A.J. (1953), "Cycling in the Simplex Algorithm", National Bureau of Standards Report 2974, December 16, 7 pages.
- Leontief, W.W. (1936), "Quantitative Input and Output Relations in the Economic System of the United States", *Review of Economic Statistics* 18, 105-125.
- Leontief, W.W. (1941), *The Structure of the American Economy, 1919-1929*, Cambridge, MA: Harvard University Press.
- Malinvaud, E. (1967), "Decentralized Procedures for Planning", in *Activity Analysis in the Theory of Growth and Planning*, E. Malinvaud and M.O.L. Bacharach (eds.), London: Macmillan.
- Mangasarian, O. (1969), *Nonlinear Programming*, New York: McGraw-Hill.
- Motzkin, T.S. (1936), *Beiträge zur Theorie der linearen Ungleichen*, Doctoral Thesis, University of Basel, Jerusalem: Azriel.

- Samuelson, P.A. (1951), "Abstract of a Theorem Concerning Substitutability in Open Leontief Models", Chapter 7 in *Activity Analysis of Production and Allocation*, T.C. Koopmans (ed.), New York: John Wiley.
- Samuelson, P.A. (1958), "Frank Knight's Theorem in Linear Programming", *Zeitschrift für National Ökonomie* 18, 310-317.
- Slater, M. (1951), "A Note on Motzkin's Transposition Theorem", *Econometrica* 19, 185-186.
- Tucker, A.W. (1956), "Dual Systems of Homogeneous Linear Relations", pp. 3-18 in in *Linear Inequalities and Related Systems*, H.W. Kuhn and A.W. Tucker (eds.), Princeton N.J.: Princeton University Press.
- Van Slyke, R.M. (1968), *Mathematical Programming and Optimal Control Theory*, Operations Research Center Report 68-21, University of California, Berkeley, July.
- von Neumann, J. (1947), "On a Maximization Problem", manuscript, Institute for Advanced Studies, Princeton University, November.
- Weymark, J.A. (1979), "A Reconciliation of Recent Results in Optimal Taxation Theory", *Journal of Public Economics* 12, 171-189.
- Woodland, A.D. (1982), *International Trade and Resource Allocation*, Amsterdam: North-Holland.

APPLIED ECONOMICS

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Chapter 11: Benchmarking and the Nonparametric Approach to Production Theory

1. Introduction

Data Envelopment Analysis or (DEA) is the term used by Charnes and Cooper (1985) and their co-workers to denote an area of analysis which is called the nonparametric approach to production theory² or the measurement of the efficiency of production³ by economists.

In section 2, we will provide an introduction to the theory of benchmarking and the measurement of relative efficiency of production units. Section 3 develops measures of relative efficiency that use only quantity data. These measures are particularly useful in the context of measuring the efficiency of government owned enterprises or units of the general government sector that deliver services to the public for free or for prices that do not reflect costs of production. Efficiency measures that use only quantity data (and not price data) are also useful in the regulatory context.⁴ Section 4 develops measures of relative efficiency for production units in the same industry where reliable price and quantity data are available for the units in the sample. Section 5 notes some relationships between the various efficiency measures developed in the previous two sections.

For applications of benchmarking to improve the efficiency of utilities or government enterprises, see Zeitsch, Lawrence and Salerian (1994), Lawrence (1995)(1998), Zeitsch and Lawrence (1996), Lawrence, Houghton and George (1997) and Swan, Lawrence and Zeitsch (2000).⁵ One very useful aspect of these benchmarking studies is that the most efficient production unit is identified by the technique so that the less efficient production units can then examine the production techniques used by the efficient unit in order to boost their own performance.

Mendoza (1989) undertook an empirical comparison of 3 different methods for measuring productivity change in the context of time series data for Canada. The 3 different methods of comparison she considered were: (i) a nonparametric or DEA method; (ii) traditional index number methods and (iii) an econometric method based on the estimation of a unit profit function.⁶ In section 6 we will compare the DEA and index number approaches to efficiency measurement using some aggregate Canadian data.

¹ The material in this chapter is drawn from Diewert and Mendoza (2007).

² See Hanoch and Rothschild (1972), Diewert (1981), Diewert and Parkan (1983) and Varian (1984). It should be noted that in recent times, the term “nonparametric approach to production theory” has sometimes included index number methods for defining the relative efficiency of production units.

³ See Farrell (1957), Afriat (1972), Färe and Lovell (1978), Färe, Grosskopf and Lovell (1985) and Coelli, Prasada Rao and Battese (1997). The last two books provide a good overview of the subject.

⁴ See Diewert (1981).

⁵ See also many of the studies in Fox (2002). Another very useful reference to DEA is the Steering Committee for the Review of Commonwealth/State Service Provision (1997). This study was largely written by Denis Lawrence.

⁶ For material on variable and unit profit functions, see Diewert (1973) (1974) and Diewert and Wales (1992). Coelli, Prasada Rao and Battese (1997) also compared the three approaches to the measurement of efficiency. Balk (1998; 179-209) also compared the three approaches. Diewert (1980) was perhaps the first to contrast the three approaches and he also included a fourth approach: the Divisia approach. The index number approach was reviewed in detail by Diewert and Nakamura (2003).

Farrell (1957; 254) noted the formal similarity of his definition of technical efficiency to Debreu's (1951) coefficient of resource utilization.

Farrell (1957; 255) also defined two further efficiency concepts using a diagram similar to Figure 1. Suppose Firm 1 faced the fixed input prices w_1 and w_2 for the two inputs. Then we could form a family of isocost lines with slope $-w_1/w_2$ and find the lowest such isocost line that is just tangent to the free disposal convex hull of the 5 points. In Figure 1, this is the line CE which is tangent to the point P^3 . Farrell noted that even if the point P^1 were shrunk in towards the origin to end up at the technically efficient point D, the resulting point would still not be the cost minimizing input combination (which is at P^3). Thus Farrell defined the *price efficiency* of P^1 as the ratio of distances OC/OD. Finally, Farrell (1957; 255) defined the *overall efficiency* of Firm 1 as the ratio of distances OC/OP¹. This measure incorporates both technical and allocative inefficiency. A point P^i is *overall efficient* if its overall efficiency is unity.

There is a problem with Farrell's measure of technical efficiency: Farrell's definition makes the points P^2 and P^5 in Figure 1 technically efficient when it seems clear that they are not: P^2 is dominated by P^3 which uses less of input 1 to produce the same output and P^5 is dominated by P^4 which uses less of input 2 to produce the same output. Charnes, Cooper and Rhodes (1978; 437) and Färe and Lovell (1978; 151) all noticed this problem with Farrell's definition of technical efficiency and suggested remedies. However, in the remainder of this chapter we will stick with Farrell's original definition of technical efficiency, with a few modifications to cover the case of many outputs.

Farrell's basic ideas outlined above for the case of a one output, constant returns to scale technology can be generalized in several ways: (i) we can relax the assumption of constant returns to scale; (ii) we can extend the analysis to the multiple output, multiple input case; (iii) we can generalize the analysis to cover situations where it is reasonable to assume profit maximizing behaviour (or partial profit maximizing behaviour) rather than cost minimizing behaviour and (iv) we can measure inefficiency in different metrics (i.e., instead of measuring technical inefficiency in terms of a proportional shrinkage of the input vector, we could choose to measure the inefficiency in terms of a basket of outputs or a basket of outputs and inputs). Drawing on the work of Mendoza (1989) and others, we shall indicate how the above generalizations (i)-(iii) can be implemented for the case of technologies that produce only 2 outputs and utilize only 2 inputs. The generalization to many outputs and inputs is straightforward. Section 3 below covers approaches that use only quantity data while section 4 describes approaches that utilize both price and quantity data. Section 5 draws on the results of the previous two sections and notes some interesting general relationships between various measures of efficiency loss. Of particular interest is a Le Chatelier Principle for measures of allocative inefficiency.

3. Efficiency Tests Using Only Quantity Data

3.1 The Case of a Convex Technology

Suppose that we have quantity data on K production units that are producing 2 outputs using 2 inputs. Let $y_m^k \geq 0$ denote the amount of output m produced by each production unit (or firm or plant) k for $m = 1, 2$, and let $x_n^k \geq 0$ denote the amount of input n used by firm k for $n = 1, 2$ and $k = 1, 2, \dots, K$.

We assume that each firm has access to the same basic technology. An approximation to the basic technology is defined to be the convex free disposal hull of the observed quantity data $\{(y_1^k, y_2^k, x_1^k, x_2^k) : k = 1, \dots, K\}$. This technology assumption is consistent

with decreasing returns to scale (and constant returns to scale) but it is *not* consistent with increasing returns to scale.

It is necessary to specify a *direction* in which possible inefficiencies are measured; i.e., do we measure the inefficiency of observation i in terms of output m or input n or some combination of outputs and inputs? Mendoza's (1989) methodology allowed for an arbitrary efficiency direction⁸ but for simplicity, we will restrict ourselves to the Debreu (1951) – Farrell (1957) direction; i.e., we shall measure the inefficiency of observation i by the smallest positive fraction δ_i^* of the i th input vector (x_1^i, x_2^i) which is such that $\{(y_1^i, y_2^i, \delta_i^* x_1^i, \delta_i^* x_2^i)\}$ is on the efficient frontier spanned by the convex free disposal hull of the K observations. If the i th observation is efficient relative to this frontier, then $\delta_i^* = 1$; the smaller δ_i^* is, then the lower is the efficiency of the i th observation. The number δ_i^* can be determined as the optimal objective function of the following linear programming problem:⁹

$$(1) \delta_i^* = \min_{\delta_i \geq 0, \lambda_1 \geq 0, \dots, \lambda_K \geq 0} \{ \delta_i : \begin{aligned} &\sum_{k=1}^K y_1^k \lambda_k \geq y_1^i; \\ &\sum_{k=1}^K y_2^k \lambda_k \geq y_2^i; \\ &\sum_{k=1}^K x_1^k \lambda_k \leq \delta_i x_1^i; \\ &\sum_{k=1}^K x_2^k \lambda_k \leq \delta_i x_2^i; \\ &\sum_{k=1}^K \lambda_k = 1 \}. \end{aligned}$$

Thus we look for a convex combination of the K data points that can produce at least the observation i combination of outputs (y_1^i, y_2^i) and use only δ_i times the observation i combination of inputs (x_1^i, x_2^i) . The smallest such δ_i is δ_i^* .

The linear programming problems (1) are run for each observation i and the resulting $\delta_i^* \geq 0$, serves to measure the relative efficiency of observation i ; if $\delta_i^* = 1$, then observation i is efficient. At least one of the K observations will be efficient.

We turn now to the corresponding linear program that tests for efficiency under the maintained hypothesis that the underlying technology is subject to constant returns to scale (in addition to being convex).

3.2 The Case of a Convex, Constant Returns to Scale Technology

In this case, the approximation to the underlying technology set is taken to be the free disposal hull of the convex cone spanned by the K data points. The efficiency of observation i is measured by the positive fraction δ_i^{**} of the i th input vector (x_1^i, x_2^i) which is such that $\{(y_1^i, y_2^i, \delta_i^{**} x_1^i, \delta_i^{**} x_2^i)\}$ is on the efficient frontier spanned by the conical convex free disposal hull of the K observations. The efficiency of the i th observation relative to this technology set can be calculated by solving the following linear program:¹⁰

⁸ See Mendoza (1989; 25-30).

⁹ See Mendoza (1989; 30) for a general version of Test 1. The use of linear programming techniques to calculate nonparametric efficiencies was first suggested by Hoffman (1957; 284) and first used by Farrell and Fieldhouse (1962). Related tests are due to Afriat (1972; 571) and Diewert and Parkan (1983; 141).

¹⁰ See Mendoza (1989; 44) for a general version of this test that allows for arbitrary directions for the i th observation to be contracted to the efficient production frontier. Here we use the Debreu (1951) proportional input direction. For related versions of this test, see Hanoch and Rothschild (1972; 268-270).

$$(2) \delta_i^{**} = \min_{\delta_i \geq 0, \lambda^1 \geq 0, \dots, \lambda^K \geq 0} \{ \delta_i \text{ subject to: } \begin{aligned} \sum_{k=1}^K y_1^k \lambda_k &\geq y_1^i; \\ \sum_{k=1}^K y_2^k \lambda_k &\geq y_2^i; \\ \sum_{k=1}^K x_1^k \lambda_k &\leq \delta_i x_1^i; \\ \sum_{k=1}^K x_2^k \lambda_k &\leq \delta_i x_2^i. \end{aligned} \}$$

Note that the LP (2) is the same as (1) except that the constraint $\sum_{k=1}^K \lambda_k = 1$ has been dropped. Thus the optimal solution for (1) is feasible for (2) and thus $\delta_i^{**} \leq \delta_i^*$; i.e., the constant returns to scale measure of efficiency for observation i will be equal to or less than the convex technology measure of inefficiency for observation i .

We turn now to models that are consistent with increasing returns to scale.

3.3 Quasiconcave Technologies

We first need to define what we mean by a production possibilities set $L(y_1)$ that is conditional on an amount y_1 of output 1. Let S be the set of feasible outputs and inputs. Then $L(y_1)$ is defined to be the set of (y_2, x_1, x_2) such that (y_1, y_2, x_1, x_2) belongs to S ; i.e., $L(y_1)$ is the set of other outputs y_2 and inputs x_1 and x_2 that are consistent with the production of y_1 units of output 1. We assume that the family of production possibilities sets $L(y_1)$ has the following three properties: (i) for each $y_1 \geq 0$, $L(y_1)$ is a closed, convex set;¹¹ (ii) if $y_1' \leq y_1''$, then $L(y_1'')$ is a subset of $L(y_1')$ and (iii) the sets $L(y_1)$ exhibit free disposal.

For each observation i , define the following set of indexes:

$$(3) I_1^i \equiv \{k : y_1^k \geq y_1^i, k = 1, 2, \dots, K\};$$

i.e., I_1^i is the set of observations k such that the amount of output 1 produced by observation k is equal to or greater than the amount of output 1 produced by observation i . Note that observation i must belong to I_1^i .

Given our assumptions on the underlying technology, it can be seen that the free disposal convex hull of the points (y_2^j, x_1^j, x_2^j) , $j \in I_1^i$, forms an approximation to the set $L(y_1^i)$. The frontier of this set is taken to be the efficient set. As usual, we measure the efficiency of observation i by the positive fraction δ_i^{***} of the i th input vector (x_1^i, x_2^i) which is such that $\{(y_2^k, \delta_i^{***} x_1^k, \delta_i^{***} x_2^k)\}$ is on the efficient frontier defined above. The number can be calculated by solving the following linear program:¹²

$$(4) \delta_i^{***} = \min_{\delta_i \geq 0, \lambda^1 \geq 0, \dots, \lambda^K \geq 0} \{ \delta_i : \begin{aligned} \sum_{k \in I_1^i} y_2^k \lambda_k &\geq y_2^i; \\ \sum_{k \in I_1^i} x_1^k \lambda_k &\leq \delta_i x_1^i; \\ \sum_{k \in I_1^i} x_2^k \lambda_k &\leq \delta_i x_2^i; \end{aligned} \}$$

and Diewert and Parkan (1983; 142). The one output version of this test is due to Farrell (1957; 258), Farrell and Fieldhouse (1962; 264) and Afriat (1972; 573).

¹¹ If we represent the underlying technology by means of the production function $y_1 = f(y_2, x_1, x_2)$, assumption (i) implies that f is a quasiconcave function.

¹² See Mendoza (1989; 54) for a general version of (4) which she called Test 3. The one output quasiconcavity test is due to Hanoch and Rothschild (1972; 259-261). Diewert (1980; 264)(1981) and Diewert and Parkan (1983; 140) developed alternative methods for dealing with a quasiconcave technology but the present method seems preferable.

$$\sum_{k \in I^i} \lambda_k = 1 \} .$$

On the left hand side of each constraint in (4), the indexes k must belong to the index set I^i defined by (3) above.

Denote the optimal λ_k for (4) above by λ_k^{***} for $k \in I^i$. By the last constraint in (4), we have

$$(5) \sum_{k \in I^i} \lambda_k^{***} = 1.$$

Using definition (3), $\lambda_k^{***} \geq 0$ and (5), it can be seen that

$$(6) \sum_{k \in I^i} y_1^k \lambda_k^{***} \geq y_1^i.$$

Using (1), (4) and (6), we see that the optimal solution for (4) is feasible for (1) and thus we must have $\delta_i^* \leq \delta_i^{***}$. Recall that we showed that $\delta_i^{**} \leq \delta_i^*$ and so we have

$$(7) 0 \leq \delta_i^{**} \leq \delta_i^* \leq \delta_i^{***}. \quad (7)$$

Thus the efficiency measures generally *increase* (or remain constant) as we make *weaker* assumptions on the underlying technology: the biggest efficiency measure δ_i^{***} corresponds to a quasiconcave (in output 1) technology, the next measure δ_i^* corresponds to a convex technology, and the smallest efficiency measure δ_i^{**} corresponds to a constant returns to scale convex technology.

In definition (3) and in the LP (4), output 1 was singled out to play a special role. Obviously, analogues to (3) and (4) could be constructed where output 2 played the asymmetric role. In this latter case, the underlying technological assumption is that the $y_2 = f(y_1, x_1, x_2)$ production function is quasiconcave. This is a somewhat different technological assumption than our initial one, but both assumptions are consistent with an increasing returns to scale technology.¹³

This completes our overview of nonparametric efficiency tests that involve the use of quantity data. We now turn to tests that involve both price and quantity data so that overall efficiency measures can be constructed in place of the technical efficiency measures of this section.

4. Efficiency Tests Using Price and Quantity Data

4.1 The Convex Technology Case

We make the same assumptions on the underlying technology as in section 3.1 above. However, we now assume that each producer may be either minimizing cost or maximizing profits.¹⁴ We consider each case in turn.

Case (i): Cost Minimization:

¹³ Mendoza's (1989; 54) Test 3 can also be modified to model quasiconcave technologies of the form $x_1 = g(y_1, y_2, x_2)$, where g is now a factor requirements function.

¹⁴ In contrast to the technical efficiency measures defined in section 2 where at least one observation had to be efficient (with an efficiency score of 1), in this section, it can be the case that no observation is efficient.

We assume that producer k faces the input prices (w_1^k, w_2^k) for the two inputs. To determine whether producer i is minimizing cost subject to our convex technology assumptions, we solve the following linear program:¹⁵

$$(8) \min_{\lambda^1 \geq 0, \dots, \lambda^K \geq 0} \left\{ w_1^i (\sum_{k=1}^K x_1^k \lambda_k) + w_2^i (\sum_{k=1}^K x_2^k \lambda_k) : \sum_{k=1}^K y_1^k \lambda_k \geq y_1^i; \right. \\ \left. \sum_{k=1}^K y_2^k \lambda_k \geq y_2^i; \right. \\ \left. \sum_{k=1}^K \lambda_k = 1 \right\}$$

$$(9) \quad \equiv \varepsilon_i^* [w_1^i x_1^i + w_2^i x_2^i].$$

The meaning of (9) is that we define the overall efficiency measure ε_i^* for observation i by equating (9) to the optimized objective function in (8). If we set $\lambda_i = 1$ and the other $\lambda_k = 0$, we have a feasible solution for (8) which yields a value of the objective function equal to $w_1^i x_1^i + w_2^i x_2^i$. Thus $0 < \varepsilon_i^* \leq 1$. The number ε_i^* can be interpreted as the fraction of (x_1^i, x_2^i) which is such that $\varepsilon_i^* (x_1^i, x_2^i)$ on the minimum cost isocost line for observation i ; i.e., ε_i^* is an analogue to the overall efficiency measure OC/OP¹ which occurred in Figure 1.

Comparing (1) and (8), it can be seen that the optimal λ_k^* solution for (1) is a feasible solution for (8) and thus:

$$(10) \quad 0 < \varepsilon_i^* \leq \delta_i^*.$$

The second inequality in (10) simply reflects the fact that overall efficiency ε_i^* is equal to or less than technical efficiency δ_i^* (recall Figure 1 again).

Case (ii): Profit Maximization:

We now assume that firm i also faces the positive output prices (p_1^i, p_2^i) for the two outputs. To determine whether producer i is maximizing profits subject to our convex technology assumptions; we solve the following linear program:¹⁶

$$(11) \max_{\lambda^1 \geq 0, \dots, \lambda^K \geq 0} \left\{ \sum_{m=1}^2 p_m^i (\sum_{k=1}^K y_m^k \lambda_k) - \sum_{n=1}^2 w_n^i (\sum_{k=1}^K x_n^k \lambda_k) : \sum_{k=1}^K \lambda_k = 1 \right\}$$

$$(12) \quad \equiv p_1^i y_1^i + p_2^i y_2^i - \alpha_i^* [w_1^i x_1^i + w_2^i x_2^i].$$

Equating (11) to (12) defines the efficiency measure α_i^* for observation i . If we set $\lambda_i = 1$ in (11) and the other $\lambda_k = 0$, we obtain a feasible value for the objective function equal to $p_1^i y_1^i + p_2^i y_2^i - [w_1^i x_1^i + w_2^i x_2^i]$. Thus $\alpha_i^* \leq 1$. If $\alpha_i^* = 1$, then observation i is efficient relative to our assumptions on the technology and relative to the hypothesis of complete profit maximization. The interpretation of α_i^* is similar to that of ε_i^* defined above by (9).

It can be seen that the optimal $\lambda_k^* \geq 0$ solution to (8) is feasible for (11). Using this fact and the inequalities in (8), we have:¹⁷

¹⁵ See Mendoza (1989; 67) for a general version of (8) which she calls Test 4.

¹⁶ This is Mendoza's (1989; 88) Test 7. It is also a special case of her Test 4. Since there is only one constraint in the problem, the solution to (11) is $\max_k \{ \sum_{m=1}^2 p_m^i y_m^k - \sum_{n=1}^2 w_n^i x_n^k ; k = 1, 2, \dots, K \}$. For related tests, see Afriat (1972; 594) for the single output case and Hanoach and Rothschild (1972; 268-270) and Diewert and Parkan (1983; 151) for the multiple output case.

$$(13) \alpha_i^* \leq \varepsilon_i^*.$$

Thus when we assume that the underlying technology set is convex and calculate the efficiency of observation i , ε_i^* , under the assumption of cost minimizing behavior and compare this efficiency level to the relative efficiency of observation i , α_i^* , calculated under the assumption of profit maximizing behavior, we find that the relative efficiency level under the profit maximizing assumption will be equal to or less than the relative efficiency level under the cost minimizing assumption.

We now turn to the corresponding linear programs that test for the efficiency of observation i under the maintained hypothesis that the underlying technology is subject to constant returns to scale.

4.2 The Convex Conical Technology Case

Case (i): Cost Minimization:

Guided by the results of section 2.2, it can be seen that all we have to do is to drop the constraint $\sum_{k=1}^K \lambda_k = 1$ from (8). The resulting optimized objective function is set equal to $\varepsilon_i^{**} [w_1^i x_1^i + w_2^i x_2^i]$. Since the new LP has one less constraint than (8), it will generally attain a smaller optimized objective function and so ε_i^{**} will generally be smaller than ε_i^* ; i.e.,

$$(14) \varepsilon_i^{**} \leq \varepsilon_i^*.$$

By comparing the new LP to (2), we can also show

$$(15) \delta_i^{**} \geq \varepsilon_i^{**}.$$

The inequality (14) shows that making *stronger* assumptions on the underlying technology tends to *decrease* the efficiency measure; i.e., the constant returns to scale measure of the efficiency of observation i , ε_i^{**} , will be equal to or less than the convex technology measure of the efficiency of observation i , ε_i^* . The inequality (15) shows that assuming cost minimizing behaviour tends to decrease the efficiency of observation i , ε_i^{**} , compared to the measure of technical efficiency that we obtained earlier for observation i , δ_i^{**} .¹⁸

Case (ii): Profit Maximization:

As in section 2.2, we could approximate the underlying technology set by the free disposal hull of the convex cone spanned by the K data points. To determine whether observation i is on the frontier of this set, we could attempt to solve the LP problem (11) after dropping the constraint $\sum_{k=1}^K \lambda_k = 1$. Unfortunately, the resulting optimal objective function is either 0 or plus infinity. Hence a different approach is required.

¹⁷ Mendoza (1989; 76-77) showed this.

¹⁸ These results and the appropriate general test may be found in Mendoza (1989; 78), which she called Test 5.

In order to obtain an operational approach, we consider a *conditional profit maximization problem* in place of the full profit maximization problem that appears in the objective function of (11); i.e., we allow firm i to maximize profits but we assume that the level of one input is *fixed* in the short run. Thus if the fixed input is input 2, to determine whether producer i is maximizing (variable) profits subject to our convex, conical technology assumptions, we solve the following linear programming problem:¹⁹

$$\begin{aligned}
 (16) \quad & \max_{\lambda_1 \geq 0, \lambda_2 \geq 0, \dots, \lambda_K \geq 0} \{ \sum_{m=1}^2 p_m^i (\sum_{k=1}^K y_m^k \lambda_k) - \sum_{n=1}^2 w_n^i (\sum_{k=1}^K x_n^k \lambda_k) : \sum_{k=1}^K x_2^k \lambda_k \leq x_2^i \} \\
 (17) \quad & = \max_k \{ [\sum_{m=1}^2 p_m^i y_m^k - (\sum_{n=1}^2 w_n^i x_n^k)] [x_2^i / x_2^k] : k = 1, 2, \dots, K \} \\
 (18) \quad & \equiv p_1^i y_1^i + p_2^i y_2^i - \alpha_i^{**} [w_1^i x_1^i + w_2^i x_2^i]
 \end{aligned}$$

where (18) serves to define the observation i efficiency measure α_i^{**} . Note that $\lambda_i = 1$ and the other $\lambda_k = 0$ is a feasible solution for (16) and this implies that $\alpha_i^{**} \leq 1$.²¹

The simple maximization problem defined by (17) can be written in the following instructive way:

$$\begin{aligned}
 (19) \quad & \max_k \{ [\sum_{m=1}^2 p_m^i y_m^k - (\sum_{n=1}^2 w_n^i x_n^k)] [x_2^i / x_2^k] : k = 1, 2, \dots, K \} \\
 & = x_2^i \max_k \{ \sum_{m=1}^2 p_m^i [y_m^k / x_2^k] - (\sum_{n=1}^2 w_n^i [x_n^k / x_2^k]) : k = 1, 2, \dots, K \}.
 \end{aligned}$$

Note that the points $[y_1^k / x_2^k, y_2^k / x_2^k, x_1^k / x_2^k, x_2^k / x_2^k] = [y_1^k / x_2^k, y_2^k / x_2^k, x_1^k / x_2^k, 1]$ are feasible output and input vectors under our constant returns to scale assumption but where the amount of input 2 is fixed at the level 1. Thus the maximization problem in (19) scales each observed output-input vector k so that the resulting scaled last input level is equal to 1 and then we take the output and input prices faced by production unit i , $[p_1^i, p_2^i, w_1^i, w_2^i]$, evaluate unit profits at these prices for each scaled output-input vector k , $p_1^i [y_1^k / x_2^k] + p_2^i [y_2^k / x_2^k] - w_1^i [x_1^k / x_2^k] - w_2^i [x_2^k / x_2^k]$, take the maximum over k of these hypothetical profits and then scale the resulting hypothetical profits by the observation i level of the “fixed” input, which is equal to x_2^i .

Comparison of (2) and (16) shows that the optimal solution to (2) generates a feasible solution for (16) and thus

$$(20) \quad \delta_i^{**} \geq \alpha_i^{**}; \quad (20)$$

i.e., the observation i technical efficiency measure δ_i^{**} is always equal to or greater than the overall observation i (conditional on input 2) profit maximization efficiency measure α_i^{**} .

Since the LP problem (16) does not simply drop the constraint $\sum_{k=1}^K \lambda_k = 1$, the single constraint in the convex technology LP problem (11), we cannot develop an inequality between the solution to (16) and the solution to (11). However, since both problems use *all* of the price and quantity data pertaining to the K observations, typically the solutions to (11) and (16) will be similar in that the efficiencies generated by these models will

¹⁹ The constraint in (16) will hold as an equality in the optimal solution. Hence the nonnegative λ_k^* which solve (16) serve to define a weighted combination of the K data points which uses the observation i amount of input 2, x_2^i , and maximizes profits at the prices of observation i .

²⁰ We require $x_2^k > 0$ for $k = 1, 2, \dots, K$ in order to derive (17) from (16).

²¹ A sufficient condition to ensure that the solution to (16) is finite is $x_2^k > 0$ for $k = 1, \dots, K$.

tend to be much lower than the technical efficiencies generated by the models presented in section 3.

4.3 The Quasiconcave Technology Case

We consider only the cost minimization case.²²

We make the same technology assumptions as were made in section 3.3. Recall the index set I_i^1 defined by (3). To determine whether producer i is minimizing cost subject to our quasiconcave technology in output 1 assumption, we solve the following linear program:

$$(21) \min_{\lambda_1 \geq 0, \lambda_2 \geq 0, \dots, \lambda_K \geq 0} \{w_1^i(\sum_{k \in I_i^1} x_1^k \lambda_k) + w_2^i(\sum_{k \in I_i^1} x_2^k \lambda_k) : \sum_{k \in I_i^1} y_2^k \lambda_k \geq y_2^i ;$$

$$\sum_{k \in I_i^1} \lambda_k = 1\}$$

$$(22) \equiv \varepsilon_i^{***} [w_1^i x_1^i + w_2^i x_2^i]$$

As usual, ε_i^{***} is our measure of overall efficiency for observation i under our present assumptions on the technology and on the producer's behaviour. Since the index i belongs to the index set I_i^1 (recall (3)), it can be seen that $\lambda_i = 1$ and the other $\lambda_k = 0$ is feasible for the LP(21) and gives rise to the feasible value for the objective function equal to $w_1^i x_1^i + w_2^i x_2^i$. Thus $\varepsilon_i^{***} \leq 1$. It is also possible to see that the optimal δ_i^{***} , λ_i^{***} solution to (4) is a feasible ε_i, λ_k solution for (21). Thus

$$(23) 0 \leq \varepsilon_i^{***} \leq \delta_i^{***} ;$$

i.e., the (quasiconcave in output 1) cost minimizing overall efficiency for observation i , ε_i^{***} , will be equal to or less than the corresponding (quasiconcave in output 1) technical efficiency loss for observation i , δ_i^{***} .

Comparing (21) with (8) and using the definition of the index set I_i^1 (recall (3)), it can be seen that the optimal $\lambda_k^{***}, \varepsilon_i^{***}$ solution for (21) is a feasible solution for (8).²³ Thus

$$(24) \varepsilon_i^{***} \geq \varepsilon_i^* ; \tag{24}$$

i.e., the observation i efficiency measure assuming a quasiconcave technology and cost minimizing behaviour ε_i^{***} will be equal to or greater than the observation i efficiency measure assuming a convex technology and cost minimizing behaviour ε_i^* .

5. Relationships between the Efficiency Measures

The inequalities derived in the previous two sections can be summarized by two rules. Note that all efficiency measures are measured in the same metric.

Rule 1: The nonparametric efficiency measures tend to *fall* as we make more restrictive technological assumptions; i.e., the quasiconcave technology efficiency measure will be equal to or greater than the corresponding convex technology efficiency measure which

²² Mendoza (1989; 83) considered more general cases in her Test 6.

²³ Using definition (3), $\lambda_k^{***} \geq 0$ and (5), it can be seen that $\sum_{k \in I_i^1} y_1^k \lambda_k^{***} \geq y_1^i$.

in turn will be equal to or greater than the corresponding convex conical technology loss measure.

Rule 2: The nonparametric efficiency measures tend to *fall* as we assume optimizing behaviour over a larger number of goods; i.e., the technical efficiency measure will be equal to or greater than the corresponding cost minimizing efficiency measure which will be equal to or greater than the corresponding profit maximizing efficiency measure. This is Mendoza's (1989; 76-77) Le Chatelier Principle for measures of allocative efficiency.

We illustrate some of the above points using some Canadian data in the following section.

6. An Empirical Comparison of Alternative Efficiency Measures for Canada

We use National Accounts and OECD data for Canada for the years 1980-2004 in order to illustrate the above programs.²⁴ Producer data on three (net) outputs and two primary inputs are used. The three net outputs are: domestic output, y_1 ($C + G + I$); exports, y_2 ; and minus imports, y_3 . The two primary inputs are: labour, x_1 and reproducible capital, x_2 . These data are listed below in Table 1.²⁵

Table 1: Quantity Data on Net Outputs and Primary Inputs for Canada, 1980-2004

Year	y_1	y_2	y_3	x_1	x_2
1980	88.22	23.23	-25.38	42.36	36.83
1981	91.73	23.62	-26.02	44.11	38.24
1982	85.45	23.20	-21.80	42.68	40.07
1983	89.07	24.63	-24.01	42.90	40.60
1984	93.38	29.21	-28.12	43.97	41.52
1985	98.49	30.63	-30.48	45.27	42.82
1986	101.71	31.97	-32.68	46.76	44.38
1987	106.53	32.95	-34.43	47.92	46.02
1988	112.55	35.95	-39.10	49.44	48.04
1989	116.91	36.24	-41.39	50.53	50.46
1990	116.22	37.97	-42.23	50.88	53.07
1991	114.12	38.66	-43.28	49.91	54.92
1992	114.44	41.45	-45.31	49.47	56.14
1993	115.95	45.97	-48.66	49.68	57.03
1994	119.46	51.83	-52.58	50.64	57.94
1995	121.39	56.22	-55.60	51.60	59.29
1996	122.85	59.40	-58.42	51.87	60.72
1997	130.60	64.35	-66.78	52.95	62.06
1998	133.68	70.18	-70.19	54.25	64.51
1999	139.19	77.75	-75.66	55.87	66.80
2000	145.42	84.61	-81.75	57.50	69.42

²⁴ We did not compute the quasiconcavity efficiencies since these tend to be close to 1 and are not very informative.

²⁵ See Diewert (2005) for a description of how the data were constructed.

2001	147.83	81.96	-77.62	58.36	72.42
2002	155.53	82.19	-78.29	59.70	74.95
2003	162.32	81.51	-82.07	60.93	77.73
2004	168.06	85.49	-88.78	61.75	80.95

The corresponding producer prices, p_1 , p_2 , p_3 for net outputs and w_1 and w_2 for primary inputs are listed in Table 2.²⁶

Table 2: Price Data on Net Outputs and Primary Inputs for Canada, 1980-2004

Year	p_1	p_2	p_3	w_1	w_2
1980	3.0783	3.7382	3.3640	4.3250	2.8210
1981	3.4053	4.0361	3.7466	4.6735	3.1366
1982	3.7361	4.1491	3.9089	5.1695	3.2346
1983	3.9537	4.1960	3.9273	5.4053	3.3299
1984	4.1081	4.3480	4.1334	5.6786	3.4856
1985	4.2730	4.4370	4.2510	5.9370	3.5477
1986	4.4630	4.4283	4.3272	6.1151	3.6578
1987	4.6241	4.5167	4.2734	6.5117	3.8049
1988	4.8124	4.5288	4.1715	6.9206	3.9791
1989	5.0277	4.6281	4.1734	7.2986	4.1243
1990	5.2515	4.5938	4.2160	7.6279	4.1099
1991	5.4192	4.4235	4.1456	8.0047	4.0010
1992	5.5112	4.5500	4.3145	8.2749	4.0053
1993	5.6198	4.7522	4.5751	8.4190	4.1214
1994	5.7082	5.0337	4.8661	8.4753	4.3004
1995	5.7797	5.3564	5.0152	8.5948	4.3944
1996	5.8433	5.3863	4.9523	8.7775	4.4662
1997	5.9309	5.3945	4.9883	9.1036	4.6262
1998	5.9969	5.3772	5.1623	9.3415	4.6238
1999	6.0794	5.4361	5.1471	9.5678	4.6415
2000	6.2151	5.7743	5.2620	10.0450	4.7601
2001	6.3336	5.8617	5.4230	10.3032	4.7135
2002	6.4492	5.7705	5.4544	10.4646	4.7970
2003	6.5617	5.6630	5.0758	10.6265	4.8548
2004	6.6784	5.7843	4.9621	10.8718	5.0059

The tests for technical efficiency of each observation, (1) and (2) in sections 3.1 and 3.2, were run using the quantity data listed in Table 1 above.²⁷ The relative technical efficiencies of the year i observation assuming a convex technology set, δ_i^* , and assuming a convex, constant returns to scale technology set, δ_i^{**} , are listed in Table 3 below. The cost minimization relative efficiencies ε_i^* defined by (8) and (9) in section 4.1 for the case of a convex technology and ε_i^{**} defined in section 4.2 for the case of a convex, constant returns to scale technology are also listed in Table 3 below. The profit maximization relative efficiencies α_i^* defined by (11) and (12) in section 4.1 for the case of a convex technology and α_i^{**} defined by (16) and (18) in section 4.2 for the case of a convex, constant returns to scale technology (with capital fixed) are also listed in Table 3 below.

²⁶ All prices were normalized to equal 1 in the year 1960. We did not use our data set which extends back to 1960 in the interests of presenting smaller tables.

²⁷ We have three (net) outputs instead of two outputs but the reader need only modify the tests in the obvious ways.

Finally, we use the data in Tables 1 and 2 to construct:

- a chained Fisher (1922) ideal index of net outputs, Y^t for year t ;
- a chained Fisher ideal index of primary inputs X^t for year t and
- a measure of *index number productivity* in year t equal to $\text{Prod}^t \equiv Y^t/X^t$.

In order to make the resulting index number estimates of Canada's productivity for the years 1980-2004, we normalize the productivities by dividing by Prod^{2002} . This makes the resulting normalized index number estimates of productivity, NProd^i , comparable to the profit maximizing estimates of relative efficiency listed in Table 3, since we had $\alpha_{2002}^* = \alpha_{2002}^{**} = 1$ and the year 2002 was the only efficient observation for both α_i^* and α_i^{**} . The normalized index number estimates of productivity are listed in the last column of Table 3.

Table 3: Relative Efficiencies for Canada, 1980-2004

Year i	δ_i^*	δ_i^{**}	ε_i^*	ε_i^{**}	α_i^*	α_i^{**}	NProd^i
1980	1.0000	1.0000	1.0000	0.9977	0.8308	0.8847	0.8629
1981	1.0000	1.0000	1.0000	1.0000	0.8480	0.8922	0.8604
1982	1.0000	1.0000	1.0000	1.0000	0.7574	0.8438	0.8422
1983	1.0000	1.0000	1.0000	1.0000	0.7659	0.8659	0.8630
1984	1.0000	1.0000	1.0000	1.0000	0.8163	0.8982	0.8894
1985	1.0000	1.0000	1.0000	1.0000	0.8345	0.9121	0.9015
1986	0.9912	0.9909	0.9893	0.9880	0.8343	0.9072	0.8929
1987	1.0000	1.0000	1.0000	1.0000	0.8465	0.9114	0.9026
1988	1.0000	1.0000	1.0000	1.0000	0.8600	0.9156	0.9095
1989	1.0000	1.0000	1.0000	1.0000	0.8528	0.9042	0.9021
1990	0.9844	0.9810	0.9728	0.9706	0.8345	0.8830	0.8833
1991	0.9824	0.9666	0.9596	0.9437	0.8170	0.8619	0.8655
1992	0.9874	0.9635	0.9601	0.9432	0.8273	0.8665	0.8717
1993	0.9890	0.9525	0.9632	0.9406	0.8457	0.8805	0.8844
1994	0.9924	0.9502	0.9732	0.9497	0.8767	0.9075	0.9088
1995	0.9882	0.9479	0.9704	0.9435	0.8804	0.9113	0.9147
1996	0.9922	0.9449	0.9701	0.9372	0.8857	0.9132	0.9179
1997	1.0000	0.9807	0.9955	0.9526	0.9147	0.9337	0.9355
1998	0.9978	0.9752	0.9892	0.9534	0.9322	0.9436	0.9457
1999	0.9992	0.9982	0.9945	0.9791	0.9580	0.9671	0.9675
2000	1.0000	1.0000	1.0000	1.0000	0.9795	0.9854	0.9838
2001	1.0000	1.0000	1.0000	1.0000	0.9780	0.9806	0.9812
2002	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2003	1.0000	1.0000	1.0000	1.0000	0.9951	0.9918	0.9926
2004	1.0000	1.0000	1.0000	1.0000	0.9985	0.9910	0.9928

Looking at Table 3, it can be seen that the various efficiency measures satisfy the following inequalities, which we showed in sections 3 and 4 must be satisfied:

$$(25) \delta_i^{**} \leq \delta_i^* ;$$

$$(26) \varepsilon_i^{**} \leq \varepsilon_i^* ;$$

$$(27) \alpha_i^* \leq \varepsilon_i^* \leq \delta_i^* ;$$

$$(28) \varepsilon_i^{**} \leq \delta_i^{**} ;$$

$$(29) \alpha_i^{**} \leq \delta_i^{**}.$$

For the Canadian data set, we also find *empirically* that

$$(30) \alpha_i^{**} \leq \varepsilon_i^{**}.$$

However, we cannot establish the inequality (30) as a *theoretical* certainty. Looking at α_i^* versus α_i^{**} , for the Canadian data, it can be seen that for the most part, $\alpha_i^* \leq \alpha_i^{**}$ and sometimes α_i^* is substantially below α_i^{**} ; i.e., the relative efficiency of an observation when we assume profit maximizing behavior and a convex technology, α_i^* , is generally less than the corresponding relative efficiency of an observation when we assume profit maximizing behavior subject to a fixed capital constraint and a convex, constant returns to scale technology, α_i^{**} . However, for the years 2003 and 2004, this relationship does not hold.

Perhaps the most interesting thing to note about the results listed in Table 3 is that with the exception of the first two years, the index number estimates of efficiency, $NProd^i$, are reasonably close to the efficiency estimates, α_i^{**} , which are based on a (variable) profit maximizing model where we assume capital is fixed and assume that there is a convex, constant returns to scale technology. These results are similar to the results obtained by Mendoza (1989; 111), who obtained nonparametric productivity indexes that were quite similar to the corresponding index number measures of productivity.²⁸

7. A Comparison of the Alternative Methods for Measuring Productive Efficiency

We summarize our comparison of alternative methods for measuring the relative efficiency of a number of production units in the same industry in point form.

- Nonparametric or DEA techniques have an overwhelming advantage over index number and econometric methods when *only* quantity data are available. Index number methods cannot be implemented without a complete set of price and quantity data. Econometric methods (i.e., production function methods) are not likely to be successful if only quantity data are available due to limited degrees of freedom.²⁹
- The relative efficiency of any single observation will tend to decrease as the sample size increases. All three methods have this problem.
- Nonparametric efficiency scores will tend to increase as we make less restrictive assumptions on the underlying technology; i.e., a quasiconcave technology set is less restrictive than a convex technology set which in turn is less restrictive than a constant returns to scale convex technology set. Econometric estimates of efficiency (technical progress estimates) may increase or decrease as we move

²⁸ Mendoza (1989; 129-134) also obtained econometric estimates of sectoral technical change for Canada and she compared these estimates with her nonparametric estimates of sectoral technical change. Her results showed that the econometric estimates of efficiency change are simply a highly smoothed version of the corresponding nonparametric estimates. Diewert and Wales (1992; 718) and Fox (1996) showed that econometric estimates of efficiency change were approximately equal to smoothed versions of index number estimates of productivity growth.

²⁹ Diewert (1992) discusses this point at some length.

from a constant returns to scale model to a nonconstant returns model. If there are increasing returns to scale, technical progress estimates will fall as we move from constant returns to scale to nonconstant returns and vice versa if there are decreasing returns to scale. Index number estimates of efficiency remain unchanged as we change our assumptions on the technology.

- Nonparametric efficiency scores will tend to decrease as we make stronger assumptions about the optimizing behaviour of producers; recall Rule 2 in section 5. It is not clear what will happen to econometric based efficiency scores under the same conditions. Since index number methods are based on the assumption of complete optimizing behaviour we cannot vary our assumptions on optimizing behaviour when using index number methods.
- If we hold the number of observations in our sample constant but disaggregate the data so that the number of inputs or outputs is increased, then nonparametric efficiency scores will tend to increase³⁰. However, index number efficiency scores will generally remain unaffected by increasing disaggregation³¹. It is not clear what will happen using econometric methods.
- The cost of computing index number estimates of relative efficiency is extremely low; the cost of the nonparametric estimates is low and the cost of computing econometric estimates can be very high if the number of goods exceeds 20 and flexible functional form techniques are used.³²
- When complete price and quantity data are available, the nonparametric estimates based on a constant returns to scale technology and profit maximizing behaviour (subject to one input being fixed) are approximately equal to the corresponding index number estimates. Econometric estimates based on the same assumptions will tend to be similar to the first two sets of estimates (but much smoother in the time series context).
- Nonparametric techniques can be adapted to deal with situations where input prices are available but not output prices. Econometric techniques can also deal with this situation but index number methods cannot be used in this situation.³³
- Nonparametric methods may be severely biased due to measurement errors; i.e., the best or most efficient observation in a DEA study may be best simply because

³⁰ As we disaggregate, the objective functions of the various linear programming problems will remain unchanged but the feasible regions for the problems become more constrained or smaller and hence the objective function minimums for the linear programming problems will become larger. Hence, the loss measures will decrease or remain constant and thus efficiency will tend to increase as we disaggregate. This point was first made by Nunamaker (1985). The profit maximization problems (11) and (16) are not affected by disaggregation.

³¹ This follows from the approximate consistency in aggregation property of superlative index number formulae like the Fisher and Törnqvist formulae; see Diewert (1978; 889 and 895).

³² The cost of estimating a fully flexible or semiflexible functional form can be high in terms of the analyst's time in doing the econometric estimation. When curvature conditions are imposed using the normalized quadratic functional form and the number of commodities are large, then in order to ensure convergence of the nonlinear regression using Shazam, it is necessary to gradually increase the rank of the substitution matrix by adding an additional rank one matrix to the already estimated substitution matrix and then rerun the model using the finishing parameter values of the previous model as starting values for the new model and so on. The procedure terminates after an iteration where the log likelihood of the model does not increase significantly.

³³ An exception occurs if there is only one output.

some output was greatly overstated or some important input was greatly understated. Index number methods are also subject to measurement errors but econometric methods may be adapted to deal with gross outliers.

Our overall conclusion is that DEA methods for measuring relative efficiency can be used profitably in a wide variety of situations when other methods are not practical or are impossible to use.

REFERENCES

- Afriat, S.N. (1972), "Efficiency Estimation of Production Function", *International Economic Review* 13, 568-598.
- Balk, B.M. (1998), *Industrial Price, Quantity and Productivity Indices*, Norwell MA: Kluwer Academic Publishers.
- Charnes, A. and W.W. Cooper (1985), "Preface to Topics in Data Envelopment Analysis", *Annals of Operations Research* 2, 59-94.
- Charnes, A., W.W. Cooper and E. Rhodes (1978), "Measuring the Efficiency of Decision Making Units", *European Journal of Operational Research* 2, 429-444.
- Coelli, T., D.S. Prasada Rao and G. Battese (1997), *An Introduction to Efficiency and Productivity Analysis*, Boston: Kluwer Academic Publishers.
- Debreu, G. (1951), "The Coefficient of Resource Utilization", *Econometrica* 19, 273-292.
- Diewert, W.E. (1973), "Functional Forms for Profit and Transformation Functions", *Journal of Economic Theory* 6, 284-316.
- Diewert, W.E. (1974), "Applications of Duality Theory," pp. 106-171 in M.D. Intriligator and D.A. Kendrick (ed.), *Frontiers of Quantitative Economics*, Vol. II, Amsterdam: North-Holland.
- Diewert, W.E. (1978), "Superlative Index Numbers and Consistency in Aggregation", *Econometrica* 46, 883-900.
- Diewert, W.E. (1980), "Capital and the Theory of Productivity Measurement", *The American Economic Review* 70, 260-267.
- Diewert, W.E. (1981), "The Theory of Total Factor Productivity Measurement in Regulated Industries", pp. 17-44 in *Productivity Measurement in Regulated Industries*, Tom Cowing and R. Stephenson (eds.), New York: Academic Press.
- Diewert, W.E. (1992), "The Measurement of Productivity", *Bulletin of Economic Research* 44, 163-198.
- Diewert, W.E. (2005), *Chapter 4: Notes on the Construction of a Data Set for an O.E.C.D. Country*, Lecture notes for Economics 594, Department of Economics, University of British Columbia, Vancouver, Canada, V6T 1Z1. May 2005. <http://www.econ.ubc.ca/diewert/594chmpg.htm>

- Diewert, W.E. and N.F. Mendoza (2007), "The Le Chatelier Principle in Data Envelopment Analysis", pp. 63-82 in *Aggregation, Efficiency, and Measurement*, Rolf Färe, Shawna Grosskopf and Daniel Primont (eds.), New York: Springer.
- Diewert, W.E. and A.O. Nakamura (2003), "Index Number Concepts, Measures and Decompositions of Productivity Growth", *Journal of Productivity Analysis* 19, 127-159.
- Diewert, W.E. and C. Parkan (1983), "Linear Programming Tests of Regularity Conditions for Production Functions," pp. 131-158 in *Quantitative Studies on Production and Prices*, W. Eichhorn, R. Henn, K. Neumann and R.W. Shephard (eds.), Vienna: Physica Verlag.
- Diewert, W.E. and T.J. Wales (1987), "Flexible Functional Forms and Global Curvature Conditions", *Econometrica* 55, 43-68.
- Diewert, W.E. and T.J. Wales (1988), "A Normalized Quadratic Semiflexible Functional Form", *Journal of Econometrics* 37, 327-342.
- Diewert, W.E. and T.J. Wales (1992), "Quadratic Spline Models for Producer's Supply and Demand Functions", *International Economic Review* 33, 705-722.
- Färe, R. and C.A.K. Lovell (1978), "Measuring the Technical Efficiency of Production", *Journal of Economic Theory* 19, 150-162.
- Färe, R., S. Grosskopf and C.A.K. Lovell (1985), *The Measurement of Efficiency of Production*, Boston: Kluwer-Nijhoff.
- Farrell, M.J. (1957), "The Measurement of Production Efficiency", *Journal of the Royal Statistical Society, Series A*, 120, 253-278.
- Farrell, M.J. and M. Fieldhouse (1962), "Estimating Efficient Production Functions under Increasing Returns to Scale", *Journal of the Royal Statistical Society, Series A*, 125, 252-267.
- Fisher, I. (1922), *The Making of Index Numbers*, Houghton-Mifflin, Boston.
- Fox, K.J. (1996), "Specification of Functional Form and the Estimation of Technical Progress", *Applied Economics* 28, 947-956.
- Fox, K.J. (ed.) (2002), *Efficiency in the Public Sector*, Boston: Kluwer Academic Publishers.
- Hanoch, G. and M. Rothschild (1972), "Testing the Assumptions of Production Theory: A Nonparametric Approach", *Journal of Political Economy* 80, 256-275.
- Hoffman, A.J. (1957), "Discussion of Mr. Farrell's Paper", *Journal of the Royal Statistical Society, Series A*, 120, p. 284.
- Lawrence, D.A. (1995), "International Comparisons of Infrastructure Performance", *Australian Journal of Public Administration* 54(3), 393-400.

- Lawrence, D. (1998), “Benchmarking Infrastructure Enterprises”, in *Utility Regulation*, Australian Competition and Consumer Commission/Public Utilities Research Centre (eds.), AGPS, Canberra.
- Lawrence, D., J. Houghton and A. George (1997), “International Comparisons of Australia’s Infrastructure Performance”, *Journal of Productivity Analysis* 8, 361–378.
- Mendoza, M.N.F. (1989), *Essays in Production Theory: Efficiency Measurement and Comparative Statistics*, Ph.D. Theses, University of British Columbia, Vancouver, Canada.
- Nunamaker, T.R. (1985), “Using Data Envelopment Analysis to Measure the Efficiency of Non-profit Organisations: A Critical Evaluation”, *Managerial and Decision Economics* 6, 58-60.
- Steering Committee for the Review of Commonwealth/State Service Provision (1997), *Data Envelopment Analysis: A Technique for Measuring the Efficiency of Government Delivery Service*, Industry Commission, Canberra: AGPS. Website: <http://www.pc.gov.au/gsp/reports/research/dea/dea.pdf>
- Swan, P., D. Lawrence and J. Zeitsch (2000), “Infrastructure: The Contribution of Benchmarking and Economic Analysis to the Reform of the Energy, Water and Transport Sectors”, in *Contemporary Economic Issues*, Volume 2, Y. Mundlak (ed.), London: MacMillan.
- Varian, H.R. (1984), “The Nonparametric Approach to Production Analysis”, *Econometrica* 52, 579-597.
- Zeitsch, J. and D. Lawrence (1996), “Decomposing Economic Inefficiency in Base Load Power Plants”, *Journal of Productivity Analysis* 7, 359-378.
- Zeitsch, J., D. Lawrence and J. Salerian (1994), “Comparing Like With Like in Productivity Studies—Apples, Oranges and Electricity”, *Economic Record* 70, 162-70.